

Relating Definitions of Computational Differential Privacy in Wider Parameter Regimes

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Abstract. The literature on computational differential privacy (CDP) has focused almost exclusively on definitions that are computational analogs of ‘pure’ $(\epsilon, 0)$ -DP. We initiate the formal study of computational versions of approximate DP, i.e. (ϵ, δ) -DP with non-negligible δ . We focus on IND-CDP and $\text{SIM}_{\forall\exists}$ -CDP and show that the hierarchy between them when $\delta > 0$ potentially differs substantially from when $\delta = 0$. In one direction, we show that for $\delta < 1$, any mechanism which is (ϵ, δ) - $\text{SIM}_{\forall\exists}$ -CDP also is (ϵ, δ) -IND-CDP, but only if ϵ is logarithmic in the security parameter. As a special case, this proves that the existing implication from $(\epsilon, 0)$ - $\text{SIM}_{\forall\exists}$ -CDP to $(\epsilon, 0)$ -IND-CDP does not hold for arbitrary ϵ , as previously claimed. Furthermore, we prove that when the parameters are the same in IND-CDP and $\text{SIM}_{\forall\exists}$ -CDP and ϵ is superlogarithmic, there exists a *natural* task that can be solved whilst satisfying $\text{SIM}_{\forall\exists}$ -CDP but which no IND-CDP mechanism can solve. This is the first separation in the CDP literature which is not due to using a task contrived specifically in order to give rise to the separation. In the other direction, we show that the techniques for establishing an implication from $(\epsilon, 0)$ -IND-CDP to $(\epsilon, 0)$ - $\text{SIM}_{\forall\exists}$ -CDP extend only to that a mechanism being (ϵ, δ) -IND-CDP implies it is also (ϵ, δ') - $\text{SIM}_{\forall\exists}$ -CDP with $\delta' > \delta$. Finally, we show that the Groce-Katz-Yerukhimovich barrier results against separations between CDP and statistical DP hold also in the setting of non-negligible δ .

Keywords: Computational Differential Privacy

1 Introduction

In 2008 and 2009, Beimel et al.[4] and Mironov et al.[41] propose a number of computational relaxations of *statistical differential privacy (SDP)*, resulting in definitions of *computational differential privacy (CDP)*.¹ For the central (‘curator’) model of DP, three definitions are proposed in [41]; IND-CDP, SIM-CDP and $\text{SIM}_{\forall\exists}$ -CDP, each by taking a different natural approach to computationally relaxing the standard definition of $(\epsilon, 0)$ -SDP. Since we refer to it recurrently, let us call the Mironov et al.[41] paper *MPRV*, after the authors. The introduction

¹ Throughout, we use the term ‘DP’ in general cases where for the argumentation it does not matter if the definition is statistical or computational.

of the definitions sparked a line of work about understanding how the definitions relate to each other and SDP [41,29,10,26,38,46,31]. Primarily two types of relations are considered in the literature; *direct implications (DI)* – whether any mechanism satisfying one definition implies that it also satisfies another, and *more expressiveness (ME)* – whether all computational tasks solvable with one definition can also be solved with another.² On a theoretical plane, the study of these relations is akin to that of computational relaxations of other statistical constructions in cryptography, where it often is the case that applications that are impossible in the statistical case become possible in the computational one [27]. The high-level motivation of analyzing the relations between DP definitions is to figure out to which degree this is the case for DP as well [27,14,41]. In multiparty models of DP it is clear that computational relaxation can increase expressiveness, due to the connection to general multiparty computation, but in the central (‘curator’) model the situation is somewhat unresolved. [41,37,26]

On a practical plane, studying the relative strength between DP definitions is important for the process of mechanism design and analysis. Say that one needs a mechanism with a specific accuracy for a given task. Then understanding differences in expressiveness between DP definitions is key to choosing which definition to use, since it answers questions such as “Must I use CDP in order to get the accuracy I need for my task? If so, which flavor of CDP must I use?”. Understanding direct implications is important when one has chosen a definition for which it is known that a good mechanism exists, but not how it looks. If one does know a mechanism for another definition then direct implications (or the lack of them) tells us if the known mechanism must be adapted in order to satisfy the DP definition one wants to use.

Since the introduction of CDP, the field of differential privacy has gone from a mostly theoretical scholarly field to being increasingly applied in practice. Notable deployments of DP include the US 2020 census[36] and systems by Apple[45], Google[22,32,35] and LinkedIn[44]. One lesson learned from the design of practical DP systems is that it is often preferable (or even needed to get acceptable utility) to use ϵ much larger than 1, contrary to commonly assumed when the theory of DP was initially developed. Similarly, it is common to use (ϵ, δ) -SDP with non-negligible δ (or another DP version such as *concentrated DP*[19,11]) due to their superior compositional properties, rather than $(\epsilon, 0)$ -SDP. Further, DP is often deployed using pseudorandom coins rather than truly random coins, yielding CDP guarantees rather than SDP ones. DP is also increasingly combined with other cryptographic techniques such as *multiparty computation (MPC)* [40,17,5,13,39], *homomorphic encryption* [2], *functional encryption* [23] or *zero-knowledge proofs* [8,6], sometimes necessitating the use of CDP instead of SDP. These developments make it important, for both theory and practice, to understand CDP not only in the setting of pure DP and small

² The types of separations are formalized in Section 3. Formalization of what it means to solve a task in this context is given in Section 2.

ε but also in wider regimes with large ε or non-negligible δ . This is not the least made clear by practice-oriented works adopting versions of *computational approximate DP* (*CapDP*) [21,5].

Aside from being necessary for achieving DP in use-cases where another computational cryptographic technique must be used, another major motivation for studying CDP is that also in the central model of DP, plausibly, it could be the case that CDP allows for solving computational tasks that cannot be solved with SDP. For instance, one can imagine that approximately computing a given function and satisfying (ε, δ) -SDP is possible only up to a given accuracy but that the same function can be approximated with a much higher accuracy whilst still satisfying some type of CDP with similar ε and δ . That is, it could be that there is a significant difference in expressiveness between SDP and CDP in the central model. A priori, the ground truth of the existence of separations in expressiveness could be of three different types:

1. *Case 1 – There are no separations.* Then no non-negligible utility/accuracy improvement can be gained by using CDP instead of SDP.
2. *Case 2 – There are only impractical separations.* Then non-negligible utility improvements can be achieved but only for ‘impractical’ edge-cases, such as when the parameters are unrealistic, the computational task is highly contrived or when the involved CDP mechanism is very inefficient.
3. *Case 3 – There are practical separations.* Then there are natural computational tasks and parameter regimes for which one can gain significant utility by using CDP.

Of course, the terms ‘practical’, ‘natural’ and ‘unrealistic’ are informal and open to debate, but should often be intuitively clear. Say, a result that would show a separation but only if ε is very small or very large (like $\varepsilon < 2^{-1000}$ or $\varepsilon > 2^{1000}$), should hardly be seen as having much of a direct relevance to real-world applications of DP. The same can likely be said if the task used to show a separation is meticulously designed solely for the purpose of showing that a separation exists. The current understanding is that there are indeed very strong separations between SDP and CDP in the central model (with essentially arbitrarily large difference in achievable utility) [10,26] and these apply for realistic parameter regimes ($\delta = 0$ and ε being any constant). Alas, both known separations are shown using tasks that are quite unnatural, as noted by the authors themselves. Therefore, as of yet, we still do not know if there exists any separation between SDP and CDP in the central model (in terms of expressiveness) that are relevant for applications of DP in the real world, and this is one of the major open questions about CDP [46].

The CDP definitions from MPRV are, as noted, all relaxations of ‘pure’ $(\varepsilon, 0)$ -SDP, alternatively $(\varepsilon, \text{negl})$ -SDP, rather than (ε, δ) -SDP with non-negligible δ . The line of theoretical works studying CDP has also remained in this setting of *computational pure DP* (*CpuDP*), leading to the relationship between the definitions now being relatively well-understood, except for the question of practical

separations. The focus on CpuDP has however also led to the understanding of CapDP being severely lacking. In this work, we start filling this gap by analyzing the extent to which some of the techniques used to relate CpuDP definitions can be extended into CapDP. Therefore, one might see it as that the main question we try to directly understand is “Does the understanding of how CDP definitions relate to each other in the pure case also apply to the approximate one?”. We focus on the relation between IND-CDP and $\text{SIM}_{\forall\exists}$ -CDP and find that whereas some techniques from CpuDP can be extended almost immediately to yield roughly analogous results for CapDP, other key results require significant technical work to be extended. This is a symptom of the approximate setting requiring quite different analytic tools, which is perhaps unsurprising since this also is the case for statistical DP.

1.1 Contributions

The contributions of this work are:

1. We discover a parameter restriction in the direct implication from $(\varepsilon, 0)$ - $\text{SIM}_{\forall\exists}$ -CDP to $(\varepsilon, 0)$ -IND-CDP (Corollary 2). It was previously claimed [41] that this implication holds without parameter restriction, but we show that it only holds when $\varepsilon \in O(\log(\kappa))$, where κ is the security parameter.
2. We observe that in CapDP, separations where both parameters are allowed to differ arbitrarily between the two definitions are impossible, since any algorithm fulfill any DP definition when $\delta = 1$. Therefore we introduce the notion of δ -gaps for quantifying relationships between CDP definitions in terms of how parameter choices in the definitions relate to each other (Section 3).
3. We prove (Corollary 2) that for any $\delta < 1$, any mechanism which is (ε, δ) - $\text{SIM}_{\forall\exists}$ -CDP also is (ε, δ) -IND-CDP with unchanged parameters, but only if ε is logarithmic in the security parameter. Our first contribution above is a special case of this result. We also show that when ε is superlogarithmic and ε, δ are the same in both definitions, there exists a *natural* task which is solvable with $\text{SIM}_{\forall\exists}$ -CDP but not with IND-CDP, being the first separation in CDP using a task not contrived specifically in order to result in a separation.
4. In the other direction, we show (Theorem 2) that the techniques for establishing a direct implication from $(\varepsilon, 0)$ -IND-CDP to $(\varepsilon, 0)$ - $\text{SIM}_{\forall\exists}$ -CDP in the approximate case only give that (for general parameters) a mechanism being (ε, δ) -IND-CDP implies that the mechanism is (ε, δ') - $\text{SIM}_{\forall\exists}$ -CDP with δ' being roughly $e^{4\varepsilon}\delta + e^{2\varepsilon}\delta + e^\varepsilon\delta$. It is left open whether such an increase in parameters is necessary for a direct implication.
5. We show (Section 6) we show that the barriers to finding expressiveness separations between SDP and CDP due to Groce-Katz-Yerukhimovich [29] also apply to CapDP. In one of the barriers, we also again discover the new parameter restriction that ε must be logarithmic.

1.2 Putting the CpuDP Contribution into Context

The hierarchy of relative strength within CpuDP is illustrated in Figure 1, with a more nuanced summary given in Section 7. Roughly, SDP directly implies SIM-CDP, which directly implies $\text{SIM}_{\forall\exists}$ -CDP which is equivalent to IND-CDP (given that $\varepsilon \in O(\log(\kappa))$, where κ is the security parameter). In the other direction, SIM-CDP is strictly more expressive than SDP if mechanisms are forced to be efficient [10] and $\text{SIM}_{\forall\exists}$ -CDP is strictly more expressive than SIM-CDP [26]³, each separation relying on specific complexity assumptions. Our first contribution above is a corrected understanding of the direct implication from $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP to $(\varepsilon, 0)$ -IND-CDP, marked by ‘*’ in Figure 1. In MPRV, this implication was claimed without restrictions on ε but we prove that it only holds when ε is logarithmic. Since this restriction was identified in MPRV for the other direction of the implication, we now know that $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP and $(\varepsilon, 0)$ -IND-CDP are equivalent if and only if ε is logarithmic. The need for ε to be logarithmic stems from that in many proofs an additive term $e^\varepsilon \text{negl}$ arise in inequalities of computational distance and the argumentation requires that this term itself is negligible. However, the definitions of CDP only guarantee that *some* negligible function negl exists. Therefore, for the argumentation to hold, one needs an ε such that $e^\varepsilon \text{negl}$ is negligible regardless of what specific function negl is. The general answer here is that ε must be logarithmic.

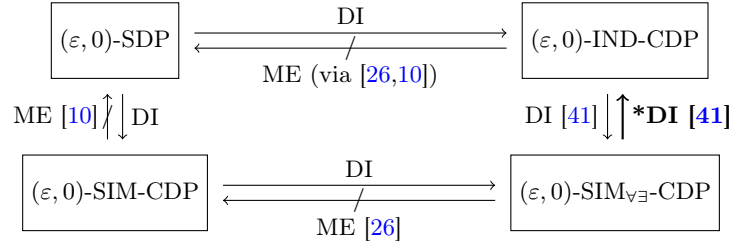


Fig. 1. Overview of relations within CpuDP in the central model, with arbitrary complexity assumptions and logarithmic ε . Some of the results apply also in wider parameter regimes (see Section 7). The abbreviations denote types of implications or separations between the definitions, for explanations, see Section 3. The separation from [10] applies only to efficient mechanisms. In the bold implication marked ‘*’, we discover a previously unknown parameter constraint (Corollary 2).

³ In [26], the separation is stated and proved between SDP and IND-CDP but, since $(\varepsilon, 0)$ -IND-CDP and $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP are equivalent in the parameter regime considered [41] and $(\varepsilon, 0)$ -SIM-CDP and $(\varepsilon, 0)$ -SDP are equally expressive up to negligible utility differences [10,26], it is clear that the separation is due to SIM-CDP and SIM $_{\forall\exists}$ -CDP being separated.

1.3 Putting the CapDP Contributions into Context

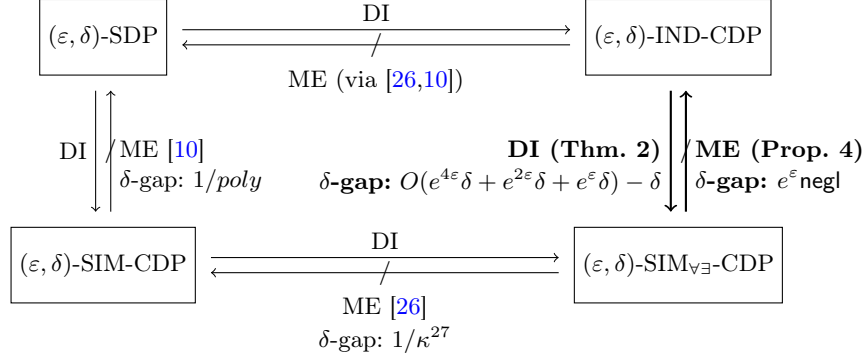


Fig. 2. Overview of relations within CapDP in the central model, with arbitrary complexity assumptions. Some of the results apply only to restricted parameter regimes (see Section 7). The separation from [10] holds only for efficient mechanisms. The bold relations are new to our work. Neither the new expressiveness separation or the new direct implication rely on any complexity assumptions.

The current understanding of the relations in CapDP is illustrated in Figure 2. Our second contribution consists of introducing the notion of δ -gaps (defined in Section 3), which for an implication is essentially how much larger the δ -parameter must be in the implied definition for the result in question to hold, when the ε 's used on both sides are the same. For a separation, the δ -gap describes how much the larger the δ -parameter can be before the separation no longer holds. Quantifying relations is necessary in CapDP since, as opposed to ε , increasing the δ -parameter always eventually make any given mechanism fulfill any given DP definition. Thus it is impossible to derive separations between any CapDP definition when the parameters are allowed to differ arbitrarily between the two definitions in question. This forces us to look for more fine-grained, or quantified, relations. Intuitively, for a separation, a larger δ -gap indicate a stronger separation, and reversed for implications.

Our third and fourth contributions are about the relations between (ε, δ) -IND-CDP and (ε, δ) -SIM $_{\forall\exists}$ -CDP. The critical difference to Figure 1 is that (ε, δ) -IND-CDP and (ε, δ) -SIM $_{\forall\exists}$ -CDP are no longer known to be equivalent even for logarithmic ε , unless δ is negligible. This is due to the new direct implication from IND-CDP to SIM $_{\forall\exists}$ -CDP having a significant δ -gap. Whether a non-zero δ -gap is necessary is left as an open problem. The implication also makes partial progress on an open problem posed in [38], asking if (ε, δ) -SIM $_{\forall\exists}$ -CDP and (ε, δ) -IND-CDP are equivalent for all $\delta > 0$ when $\varepsilon \in O(\log(\kappa))$, where the question would be closed in the negative if one can show that a non-zero δ -gap is needed. Further, when ε is superlogarithmic, (ε, δ) -SIM $_{\forall\exists}$ -CDP is strictly

more expressive than (ε, δ) -IND-CDP, in the sense that the δ parameter in IND-CDP must be increased in order to allow solving all tasks that can be solved by some (ε, δ) -SIM $_{\forall\exists}$ -CDP mechanism. This might be surprising, because one typically interprets SIM $_{\forall\exists}$ -CDP as an at least not more relaxed definition than IND-CDP [38,10], but the separation suggests that this view is not strictly true for large (superlogarithmic) ε .

Notably, this separation is for a natural task, namely that of evaluating a noisy predicate with a certain constant probability of the noise being 0, and the separation is proven by using a version of the well-known Randomized Response mechanism [47]. Our expressiveness separation is the first separation in CDP (including those for CpuDP) that follows from a natural task, since the tasks in [10,26] are contrived specifically to yield a separation. As opposed to the previous separations, however, ours only take effect when ε is superlogarithmic in the security parameter, which is arguably not a very realistic parameter regime to use in practice. Therefore, we do not think of our results as having shown the existence of practically relevant separations within CDP. Rather, our separation may be seen as suggesting an alternative route to explore the plausibility of practical separations, since it implies that separations can currently be found by *either* using natural tasks or by using natural parameter regimes but not both simultaneously. Aside from their more practical parameter regimes, the separations by [10,26] have several other advantages over ours. Firstly, their separations apply when the ε -parameters in the two DP definitions are arbitrarily different and independent from each other, where our separation is formulated with the same ε in both DP definitions (the rationale of this is explained above). Secondly, their separations exhibit *arbitrary* differences in utility between the more and the less expressive definition, whereas our separation results in a bounded utility gap dependent on ε and δ . In other words, how *much* more expressive SIM $_{\forall\exists}$ -CDP is than IND-CDP is dependent on ε and δ and, particularly, the gap is negligible if ε is logarithmic.

Our fourth contribution is adapting the techniques from MPRV for showing that $(\varepsilon, 0)$ -IND-CDP directly implies $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP. The technique entails expressing IND-CDP and SIM $_{\forall\exists}$ -CDP in terms of computational versions of density, then applying an extended *dense model theorem (DMT)* [41,43,28,3] to show that the IND-CDP mechanism $\mathcal{M}$ is, in a sense, at a bounded computational distance from a mechanism $\mathcal{S}$ that is essentially $(\varepsilon, 0)$ -SDP, in the way needed for SIM $_{\forall\exists}$ -CDP. The complication that arise in the CapDP case is, roughly, that the distance between $\mathcal{M}$ and $\mathcal{S}$ is computational rather than statistical. We overcome this obstacle by using a corollary of the Hybrid Indistinguishability Lemma [9]. The corollary says, essentially, that one can turn computational distance into statistical distance by introducing a negligible computational slack. This turns out to be enough to prove a direct implication from (ε, δ) -IND-CDP to (ε, δ') -SIM $_{\forall\exists}$ -CDP for arbitrary ε and δ but with $\delta' > \delta$.

1.4 Putting the Barrier Contribution into Context

Our fifth contribution concerns the two barrier results of [29]. The first barrier says essentially that for any computational task, an expressiveness separation between SDP and IND-CDP necessitates using stronger cryptographic tools than the black-box use of one-way functions. The second barrier says, roughly, that no matter what complexity assumption one makes, there are no expressiveness separations for tasks where the utility is measured as a bounded L_p -norm. In Section 6, we show that the proofs for these results more or less do not use the fact that δ is 0 and thus the barriers are easily extended to the setting of larger δ . The extension of the barriers to the approximate case leaves them qualitatively unchanged, meaning that there is no significant blow-up in parameters or similar which would suggest that circumventing them would be any easier in the approximate case. Further, we also point out that the proof of the second barrier relies on ε being logarithmic in the case of CpuDP. This is for the same reason that this restriction arise in the other results we treat in this work. That the barriers extend to CapDP suggests that finding a practically relevant expressiveness separation between SDP and CDP might not be significantly easier if one allows a non-negligible (but still practically meaningful) δ .

2 Preliminaries

2.1 Statistical Differential Privacy

Definition 1 ((ε, δ) -SDP [18, 16]). *Let $\varepsilon \geq 0, \delta \in [0, 1]$ and $\mathcal{D}, \mathcal{R}$ be discrete sets. A probabilistic algorithm $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ (called a mechanism) is statistically (ε, δ) -differentially private (SDP) if for all pairs (D, D') of adjacent databases (i.e. that differ in at most one element) in $\mathcal{D}$ and all subsets S of $\mathcal{R}$,*

$$\mathbb{P}(\mathcal{M}(D) \in S) \leq e^\varepsilon \mathbb{P}(\mathcal{M}(D') \in S) + \delta, \quad (1)$$

where the probability is over $\mathcal{M}$'s internal coin tosses.

Since we take an asymptotic viewpoint on mechanisms (which is necessary when dealing with CDP), we consider ensembles of mechanism indexed by the security parameter κ rather than single mechanisms. A *mechanism ensemble* is a sequence $\mathcal{M} = \{\mathcal{M}_\kappa\}_{\kappa \in \mathbb{N}}$ with $\mathcal{M}_\kappa : \mathcal{D}_\kappa \rightarrow \mathcal{R}_\kappa$. In our analyses, we typically abuse notation and refer to mechanism ensembles simply as mechanisms. Similarly, we let the privacy parameters depend on κ and thus get parameter ensembles $\varepsilon = \{\varepsilon_\kappa\}_{\kappa \in \mathbb{N}}, \delta = \{\delta_\kappa\}_{\kappa \in \mathbb{N}}$. We say that a mechanism ensemble $\mathcal{M}$ is (ε, δ) -SDP (and similarly for the CDP definitions) if for all sufficiently large κ , $\mathcal{M}_\kappa$ is $(\varepsilon_\kappa, \delta_\kappa)$ -SDP. There are many ways of defining what adjacency between databases means (see, for instance, [15, 38]) but this choice has no effect on our results.

2.2 Distances

We denote an arbitrary negligible function by negl . A function $f : \mathbb{N} \rightarrow \mathbb{R}$ is called negligible if for every polynomial p and large enough $n \in \mathbb{N}$, $f(n) < 1/p(n)$. A function f is called noticeable if there exists a polynomial p such that for large enough n , $f(n) > 1/p(n)$. Throughout, we consider ε, δ parameters that are monotone and therefore if δ is not negligible then it is noticeable. We use SD to denote statistical distance (total-variation distance) between two probability distributions and we use CD to denote non-uniform computational distance of the same, i.e. with respect to $\mathcal{T}_{poly}$, the class of all predicates over $\mathcal{R}$ computable by some non-uniform polynomial-time Turing Machine (TM). Of course, when speaking of non-uniform TMs (or algorithms) with some given runtime bound, we also restrict the advice length in the same way, as is standard.⁴ For probability ensembles $X = \{X_\kappa\}_{\kappa \in \mathbb{N}}, Y = \{Y_\kappa\}_{\kappa \in \mathbb{N}}$ over a sequence $\mathcal{R} = \{\mathcal{R}_\kappa\}_{\kappa \in \mathbb{N}}$ of sample spaces, we say that they have computational (respectively statistical) distance δ if for all κ , the computational (respectively statistical) distance between X_κ and Y_κ is no larger than δ_κ . Throughout, we assume that all elements in $\mathcal{D}_\kappa$ and $\mathcal{R}_\kappa$ have polynomial size in κ . At times we consider computational distance with respect to some other set $\mathcal{T}$ of predicate tests, which we then denote $CD_{\mathcal{T}}$. We say that two distributions X, Y are δ -indistinguishable with respect to $\mathcal{T}$ if $CD_{\mathcal{T}}(X, Y) \leq \delta$. As usual, we say that they are indistinguishable if they are negl -indistinguishable with respect to $\mathcal{T}_{poly}$.

2.3 Pseudodensity, Tests and Dense Models

Differential privacy is in essence a requirement that the output distribution of an algorithm does not change much when its input changes only a little. This requires the use of a definition of similarity between probability distributions. In the case of $(\varepsilon, 0)$ -SDP, the similarity notion is that of mutual density.

Definition 2 (Density of distributions). *Let X, Y be probability distributions on a finite universe $\mathcal{R}$ and $\beta > 0$. We say that X is β -dense in Y if*

$$\forall x \in \mathcal{R} : \beta \mathbb{P}(X = x) \leq \mathbb{P}(Y = x). \quad (2)$$

If X and Y are each β -dense in the other, we say that they are mutually β -dense.

When β is at most 1, X being β -dense in Y is, informally, equivalent to that X can be sampled by sampling from Y with probability β and otherwise

⁴ In the literature, one often switch between describing the complexity of a set of predicate tests in terms of non-uniform TMs and circuit families, depending on what is more common or convenient in the context of the tools used. For instance, in MPRV the CDP definitions are phrased in terms of non-uniform TMs (perhaps because the definitions are to be seen as primarily cryptographic) but in the section making heavy use of the dense model theorem, the language of circuit families is used (perhaps due to the techniques stemming from complexity theory). In our work we stick to TMs, for consistency, but of course the choice does not matter.

sampling from some other distribution [43]. The computational relaxation from SDP to CDP is analog to a relaxation from density to a definition of a distribution *appearing* dense in another, under some collection of predicates, which we refer to as *tests*. That is, a test is simply a predicate on $\mathcal{R}$. For CDP, the collection of tests is that of all non-uniform polynomial-time tests, which we denote $\mathcal{T}_{poly}$. Similarly, for any positive function $s : \mathbb{N} \rightarrow \mathbb{N}$, $\mathcal{T}_s$ is the class of all tests computable in time at most s . Note that we restrict ourselves to predicate tests to simplify our notation but that the theory of computational density extends also to bounded real-valued tests. Similarly, we make the restriction to deterministic predicates for simplicity but this is without loss of generality since indistinguishability (and pseudodensity), with respect to the test collections we consider (either all polynomial-time tests or of some superpolynomial runtime) remain qualitatively unchanged by allowing probabilism.⁵ We consider two definitions of a distribution appearing dense in another distribution with respect to a collection of tests, namely *pseudodensity* and that of *having a dense model*.

Definition 3 (Pseudodensity and (β, α) -density). *For distributions X, Y over $\mathcal{R}$, X is (β, α) -pseudodense in Y with respect to $\mathcal{T}$ if for all $T \in \mathcal{T}$:*

$$\mathbb{P}(T(X) = 1) - \frac{1}{\beta} \mathbb{P}(T(Y) = 1) \leq \alpha. \quad (3)$$

When $\mathcal{T}$ is not specified then we mean $\mathcal{T}_{poly}$. When X and Y are each (β, α) -pseudodense in the other, they are mutually (β, α) -pseudodense. If $\mathcal{T} = \{T : \mathcal{R} \rightarrow \{0, 1\}\}$, we say that X is (β, α) -dense in Y .

Remark 1. As noted in MPRV, a mechanism ensemble $\mathcal{M}$ is $(\varepsilon, 0)$ -SDP iff for all adjacent inputs D, D' , $\mathcal{M}_\kappa(D)$ and $\mathcal{M}_\kappa(D')$ are mutually $e^{-\varepsilon\kappa}$ -dense. Similarly, a mechanism ensemble $\mathcal{M}$ is (ε, δ) -SDP iff for all adjacent inputs D, D' and κ , $\mathcal{M}_\kappa(D)$ and $\mathcal{M}_\kappa(D')$ are mutually $(e^{-\varepsilon\kappa}, \delta_\kappa)$ -dense.

Definition 4 (Dense model). *We say that X has a β -dense α -model in Y with respect to $\mathcal{T}$ if there exists a distribution Z on $\mathcal{R}$ such that Z is β -dense in Y and X is α -indistinguishable from Z with respect to $\mathcal{T}$.*

It is immediate from the definitions that having a β -dense α -model in a distribution with respect to a set of tests implies being (β, α) -pseudodense in the same distribution with respect to the same set of tests. In the other direction, the implication does not always hold, meaning that having (β, α) -pseudodensity does not always mean having a β -dense α -model. It is however the case that for

⁵ Indistinguishability with respect to deterministic non-uniform polynomial-time TMs implies the same indistinguishability with respect to probabilistic non-uniform polynomial-time TMs [27]. Similarly, indistinguishability with respect to all deterministic non-uniform tests with some fixed superpolynomial runtime bound implies the same for all probabilistic non-uniform tests with some (other) superpolynomial runtime bound (via the same ‘coin-fix’ argument). For our results, it does not matter which superpolynomial bound is used, but rather that *some* such bound exists.

a very broad class of test collections, pseudodensity does indeed in general imply the existence of a dense model, albeit with respect to a slightly weaker set of tests and with slightly decayed indistinguishability parameter. Such results are referred to as *dense model theorems (DMTs)* [43,41,33,3].

3 Relating DP Definitions

When comparing DP definitions there are a few different types of relations that are interesting. The most basic is *equivalence*, when any mechanism fulfilling the one definition also fulfills the other, potentially with some shift in parameters. In each direction, we refer to such relations as *direct implications*, that is, we say that DP definition Def_1 directly implies DP definition Def_2 if any mechanism satisfying the first also satisfies the second. Another relation is that of *equal expressiveness*, meaning that any *task* solvable whilst satisfying one of the DP definition is also solvable whilst satisfying the other, potentially by a different mechanism. The one-directional version of this we call *more expressiveness* and when definition Def_1 is more expressive than definition Def_2 but not the other way around, then we say that definition Def_1 is *strictly more expressive* than definition Def_2 . The definitions we use are from [38], which are inspired by those in [15].⁶

Definition 5 (Direct implication (DI) [38]). *Let $\text{Def}_1, \text{Def}_2$ be elements in $\{\text{SDP}, \text{IND-CDP}, \text{SIM}_{\forall\exists}\text{-CDP}, \text{SIM-CDP}\}$ and $\varepsilon, \delta, \varepsilon', \delta'$ be parameter ensembles for them, respectively. We say that $(\varepsilon, \delta)\text{-Def}_1$ directly implies (DI) $(\varepsilon', \delta')\text{-Def}_2$, denoted $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(DI)} (\varepsilon', \delta')\text{-Def}_2$, if all mechanisms $\mathcal{M}$ satisfying $(\varepsilon, \delta)\text{-Def}_1$ also satisfy $(\varepsilon', \delta')\text{-Def}_2$. If for all fixed ε, δ there exists ε', δ' such that $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(DI)} (\varepsilon', \delta')\text{-Def}_2$ and $(\varepsilon', \delta')\text{-Def}_2 \xRightarrow{(DI)} (\varepsilon, \delta)\text{-Def}_1$, the definitions are equivalent.*

A *task* is defined with respect to a *utility function*, which is an efficient deterministic predicate mapping a dataset and a mechanism output to 1 if and only if the output was a ‘good’ solution. Since DP mechanisms are probabilistic, we measure the utility as the probability that a mechanism will output a good solution. A task is a utility function together with a required probability of success.

Definition 6 (Utility function [10,26]). *A utility function is an efficiently (uniform polynomial time) computable function $u : \mathcal{D} \times \mathcal{R} \rightarrow \{0, 1\}$. A mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is called α -useful for u if for every $D \in \mathcal{D}$:*

$$\mathbb{P}_{y \leftarrow \mathcal{M}(D)}(u(D, y) = 1) \geq \alpha. \quad (4)$$

⁶ An alternative formulation could be to think of direct implications as *mechanism implications* and their negations as *mechanism separations*, since they describe how the definitions relate on a mechanism level. Analogously, results about more expressiveness may be seen as *task implications* and *task separations*, respectively.

Definition 7 (Task). A task is a tuple $\tau = (\alpha, u)$ as in Definition 6. For a mechanism $\mathcal{M}$, we say that $\mathcal{M}$ solves the task τ if $\mathcal{M}$ is α -useful for u . With fixed ε, δ and Def as above, τ is said to be solvable whilst satisfying (ε, δ) -Def if there exists a mechanism which satisfies (ε, δ) -Def and solves τ .

For ensembles of mechanisms, we allow (just as for privacy parameters) the utility function u , utility limit α and task τ to depend on κ and then usefulness is considered in an asymptotic sense.

Definition 8 (More expressive (ME) [38]). Let $\text{Def}_1, \text{Def}_2$ be elements in $\{\text{SDP}, \text{IND-CDP}, \text{SIM}_{\forall\exists}\text{-CDP}, \text{SIM-CDP}\}$ and $\varepsilon, \delta, \varepsilon', \delta'$ be parameter ensembles for them, respectively. We say that (ε', δ') -Def₂ is more expressive (ME) than (ε, δ) -Def₁, denoted $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(ME)} (\varepsilon', \delta')\text{-Def}_2$, if all tasks solvable whilst satisfying $(\varepsilon, \delta)\text{-Def}_1$ also are solvable whilst satisfying $(\varepsilon', \delta')\text{-Def}_2$. If $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(ME)} (\varepsilon', \delta')\text{-Def}_2$ and $(\varepsilon, \delta)\text{-Def}_1 \xleftarrow{(ME)} (\varepsilon', \delta')\text{-Def}_2$ then we say that the definitions are equally expressive. If $(\varepsilon', \delta')\text{-Def}_2$ is more expressive than $(\varepsilon, \delta)\text{-Def}_1$ but not the other way around, then we say that $(\varepsilon', \delta')\text{-Def}_2$ is strictly more expressive than $(\varepsilon, \delta)\text{-Def}_1$.

From the definitions follows that DI-results imply ME-results.

Proposition 1 ([38]). For any $\varepsilon, \delta, \varepsilon', \delta', \text{Def}_1, \text{Def}_2$ as above, if $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(DI)} (\varepsilon', \delta')\text{-Def}_2$ then $(\varepsilon, \delta)\text{-Def}_1 \xRightarrow{(ME)} (\varepsilon', \delta')\text{-Def}_2$.

Note that it is possible that two DP definitions are incomparable with respect to either direct implications or expressiveness, meaning that in some sense they are well suited for different types of tasks.

3.1 Parameter Dependence in Relations and Parameter Gaps

When showing implications or separations (i.e. that one definition does not imply another in the sense of DI or ME) between definitions it is important to consider the relation between the parameters used on each side. For instance, all direct implications in the literature on CpuDP use the same ε for both of the definitions, such as in the result in MPRV that $(\varepsilon, 0)\text{-IND-CDP}$ directly implies $(\varepsilon, 0)\text{-SIM}_{\forall\exists}\text{-CDP}$ for all logarithmic ε . This means that, for every fixed such ε , any mechanism that is $(\varepsilon, 0)\text{-IND-CDP}$ also is $(\varepsilon, 0)\text{-SIM}_{\forall\exists}\text{-CDP}$, importantly *without* a change in parameters. Similarly, both known separations [10, 26] allow the two ε 's be arbitrarily different. For example does [26] show that for every constant $\varepsilon > 0$, there is a task that can be solved whilst satisfying $(\varepsilon, 0)\text{-IND-CDP}$ but that cannot be solved whilst satisfying $(\varepsilon', 0)\text{-SDP}$ for *any* constant ε' .

For CapDP, the situation is different. One reason for this is that all the DP definitions we consider become trivially satisfied if the parameters are made large enough (see Section 4.2). In particular, all algorithms are $(0, 1)\text{-SDP}$. Therefore,

in CapDP, there cannot be separations where both pairs of ε, δ are allowed to differ arbitrarily. A similar thing happens for implications, where it is possible to show that one definition directly implies another, but only under the condition that the parameters must be worse in the implied one. This begs the question of how one should interpret such results, a question for which there is not necessarily only one good answer. We suggest quantifying the implications by how much larger δ must be in the implied definition (when the ε 's are the same), compared to the one implying it, and we refer to this as the δ -gap. We suggest using the same quantification for separations, where the δ -gap is, intuitively, how much larger the δ can be made in the weaker definition before the separation in question no longer holds.

Definition 9 (δ -gap). *Let $\text{Def}_1, \text{Def}_2$ be as Definition 5. For every $\varepsilon > 0, \delta \in [0, 1]$, and $\delta' = f(\varepsilon, \delta)$, where f is a non-negative real-valued function:*

- *If (ε, δ) -Def₁ directly implies (ε, δ') -Def₂ then the δ -gap of this implication is defined as $\delta' - \delta$.*
- *If (ε, δ') -Def₂ is more expressive than (ε, δ) -Def₁ then the δ -gap of this implication is defined as $\delta' - \delta$.*
- *If (ε, δ) -Def₁ does not directly imply (ε, δ') -Def₂, then the δ -gap of this separation is defined as $\delta' - \delta$.*
- *If (ε, δ') -Def₂ is not more expressive than (ε, δ) -Def₁, then the δ -gap of this separation is defined as $\delta' - \delta$.*

Note that the δ -gap is a property of the implication itself (i.e. the theorem proving how the definitions are related) rather than of the definitions. Therefore, it could be the case that the δ -gap of the best implication from one definition to another does not match the δ -gap of the strongest separation between them.

4 Adapting the CDP Definitions to Approximate DP

The two CDP definitions from MPRV [41] that are most commonly used are IND-CDP and SIM-CDP. IND-CDP is the same as SDP for mechanism ensembles but with the two differences that the inequality is with respect to efficient tests rather than all subsets of the range and that there is an extra negligible additive term. Since the inequality in ε -IND-CDP is precisely that in $(\varepsilon, 0)$ -SDP except for the aforementioned changes, a natural way of extending IND-CDP into CapDP is to allow the additive term to be non-negligible. This is done, for instance, in [20,30].

Definition 10 ((ε, δ) -IND-CDP). *An ensemble $\mathcal{M}$ of mechanisms $\mathcal{M}_\kappa : \mathcal{D}_\kappa \rightarrow \mathcal{R}_\kappa$ is (ε, δ) -IND-CDP if there exists a negligible function negl such that for every test $T \in \mathcal{T}_{\text{poly}}$, every sufficiently large κ and all adjacent $D, D' \in \mathcal{D}_\kappa$ of polynomial size in κ , it holds that*

$$\mathbb{P}(T(\mathcal{M}_\kappa(D)) = 1) \leq e^{\varepsilon\kappa} \mathbb{P}(T(\mathcal{M}_\kappa(D')) = 1) + \delta_\kappa + \text{negl}(\kappa),$$

with the probability being over the randomness in $\mathcal{M}_\kappa$.

SIM-CDP is formalized as there existing some ‘simulator’ which is SDP and computationally indistinguishable from the mechanism. Arguably the most intuitive way of adapting SIM-CDP to CapDP is to exchange the requirement that the simulator $\mathcal{S}$ is $(\varepsilon, 0)$ -SDP with that it must be (ε, δ) -SDP. This is done, for instance, in [38].

Definition 11 ((ε, δ)-SIM-CDP). *An ensemble $\mathcal{M}$ of mechanisms $\mathcal{M}_\kappa : \mathcal{D}_\kappa \rightarrow \mathcal{R}_\kappa$ is (ε, δ) -SIM-CDP if there exists a negligible function negl and an ensemble $\{\mathcal{S}_\kappa(\cdot)\}_{\kappa \in \mathbb{N}}$ of (ε, δ) -SDP mechanisms $\mathcal{S}_\kappa : \mathcal{D}_\kappa \rightarrow \mathcal{R}_\kappa$ such that for every sufficiently large κ and every $D \in \mathcal{D}_\kappa$ of polynomial size in κ , it holds that $\mathcal{M}_\kappa(D)$ and $\mathcal{S}_\kappa(D)$ are $\text{negl}(\kappa)$ -indistinguishable with respect to $\mathcal{T}_{\text{poly}}$.*

A relaxed version of SIM-CDP is also introduced in MPRV, which differ from SIM-CDP solely in that the quantifiers are swapped. We analogously define (ε, δ) -SIM $_{\forall\exists}$ -CDP with respect to (ε, δ) -SIM-CDP. To the best of our knowledge, there is no work studying (ε, δ) -SIM $_{\forall\exists}$ -CDP in the literature.

Definition 12 ((ε, δ)-SIM $_{\forall\exists}$ -CDP). *An ensemble $\mathcal{M}$ of mechanisms $\mathcal{M}_\kappa : \mathcal{D}_\kappa \rightarrow \mathcal{R}_\kappa$ is (ε, δ) -SIM $_{\forall\exists}$ -CDP if there exists a negligible function negl such that for every adjacent $D, D' \in \mathcal{D}_\kappa$ of polynomial size in κ there exists distribution ensembles $\{\mathcal{S}_\kappa(D)\}_{\kappa \in \mathbb{N}}$ and $\{\mathcal{S}_\kappa(D')\}_{\kappa \in \mathbb{N}}$ such that for large enough κ :*

1. For all $S \subset \mathcal{R}_\kappa$:

$$\begin{aligned}\mathbb{P}(\mathcal{S}_\kappa(D) \in S) &\leq e^{\varepsilon\kappa} \mathbb{P}(\mathcal{S}_\kappa(D') \in S) + \delta_\kappa, \\ \mathbb{P}(\mathcal{S}_\kappa(D') \in S) &\leq e^{\varepsilon\kappa} \mathbb{P}(\mathcal{S}_\kappa(D) \in S) + \delta_\kappa.\end{aligned}$$

2. $\mathcal{S}_\kappa(D)$ is $\text{negl}(\kappa)$ -indistinguishable from $\mathcal{M}_\kappa(D)$ with respect to $\mathcal{T}_{\text{poly}}$ and analogously for $\mathcal{S}_\kappa(D')$ and $\mathcal{M}_\kappa(D')$.

In MPRV, SIM $_{\forall\exists}$ -CDP is introduced in order to relate SIM-CDP and IND-CDP and it is shown that $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP and $(\varepsilon, 0)$ -IND-CDP are equivalent when ε is logarithmic.

4.1 Characterizing CDP by Computational Density

Each of the CDP definitions can be characterized by different types of computational density, as noted in MPRV. In CpuDP, we have:

- A mechanism $\mathcal{M}$ is $(\varepsilon, 0)$ -IND-CDP iff for all adjacent D, D' and all sufficiently large κ , $\mathcal{M}_\kappa(D)$ and $\mathcal{M}_\kappa(D')$ are mutually $(e^{-\varepsilon\kappa}, \text{negl})$ -pseudodense with respect to $\mathcal{T}_{\text{poly}}$.
- A mechanism $\mathcal{M}$ is $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP iff for all adjacent D, D' and all sufficiently large κ , there exists distributions $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$ such that $\mathcal{S}_\kappa(D)$ is a $e^{-\varepsilon\kappa}$ -dense $\text{negl}(\kappa)$ -model of $\mathcal{M}_\kappa(D)$ in $\mathcal{S}_\kappa(D')$ and $\mathcal{S}_\kappa(D')$ is a $e^{-\varepsilon\kappa}$ -dense $\text{negl}(\kappa)$ -model of $\mathcal{M}_\kappa(D')$ in $\mathcal{S}_\kappa(D)$.

- A mechanism $\mathcal{M}$ is $(\varepsilon, 0)$ -SIM-CDP iff the same as for $\text{SIM}_{\forall\exists}$ -CDP holds but with the additional requirement that S is a fixed mechanism ensemble mapping $\mathcal{D}$ to $\mathcal{R}$.

For CapDP, we instead have:

- A mechanism $\mathcal{M}$ is (ε, δ) -IND-CDP iff for all adjacent D, D' and all sufficiently large κ , $\mathcal{M}_\kappa(D)$ and $\mathcal{M}_\kappa(D')$ are mutually $(e^{-\varepsilon\kappa}, \delta_\kappa + \text{negl}(\kappa))$ -pseudodense with respect to $\mathcal{T}_{poly}$.
- A mechanism $\mathcal{M}$ is (ε, δ) - $\text{SIM}_{\forall\exists}$ -CDP iff for all adjacent D, D' and all sufficiently large κ , there exists distributions $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$ such that $\mathcal{S}_\kappa(D)$ is a $(e^{-\varepsilon\kappa}, \delta_\kappa)$ -dense $\text{negl}(\kappa)$ -model of $\mathcal{M}_\kappa(D)$ in $\mathcal{S}_\kappa(D')$ and $\mathcal{S}_\kappa(D')$ is a $(e^{\varepsilon\kappa}, \delta_\kappa)$ -dense $\text{negl}(\kappa)$ -model of $\mathcal{M}_\kappa(D')$ in $\mathcal{S}_\kappa(D)$.
- A mechanism $\mathcal{M}$ is (ε, δ) -SIM-CDP iff the same as for $\text{SIM}_{\forall\exists}$ -CDP holds but with the additional requirement that S is a fixed mechanism ensemble mapping $\mathcal{D}$ to $\mathcal{R}$.

4.2 Trivial Relations

Directly from the definitions follows that, for all ε and δ , (ε, δ) -SDP directly implies (ε, δ) -SIM-CDP. Similarly, (ε, δ) -SDP directly implies (ε, δ) -IND-CDP and (ε, δ) -SIM-CDP directly implies (ε, δ) - $\text{SIM}_{\forall\exists}$ -CDP. The other known relations between SDP and the three CDP versions are non-trivial, with the exception of situations when the δ 's are one. These equivalences below all give rise to direct implications with trivial δ -gaps equal to 1.

Proposition 2. $(\varepsilon, 1)$ -SDP, $(\varepsilon', 1)$ -SIM-CDP, $(\varepsilon'', 1)$ - $\text{SIM}_{\forall\exists}$ -CDP and $(\varepsilon''', 1)$ -IND-CDP are all equivalent for any values of $\varepsilon, \varepsilon', \varepsilon'', \varepsilon''' \geq 0$.

Proof. Any mechanism fulfills all of them. In particular, all algorithms mapping $\mathcal{D}$ to $\mathcal{R}$ are $(0, 1)$ -SDP since $(0, 1)$ -SDP is exactly the same as upper bounding $|\mathbb{P}(\mathcal{M}(D) \in S) - \mathbb{P}(\mathcal{M}(D') \in S)|$ by 1, which already follows due to the two terms being probabilities. Further, since $(0, 1)$ -SDP implies $(\varepsilon, 1)$ -SDP for all values of ε and (ε, δ) -SDP implies both (ε, δ) -SIM-CDP and (ε, δ) -IND-CDP for all ε, δ , the proposition follows. $\square$

5 Relation Between IND-CDP and $\text{SIM}_{\forall\exists}$ -CDP

5.1 Intermezzo – Statistical Approximate DP vs Statistical Distance to Pure DP

A common interpretation of (ε, δ) -SDP is that it is essentially the same as the mechanism being $(\varepsilon, 0)$ -SDP with probability at least $1 - \delta$ [46]. These two requirements are not entirely equivalent, rather, being $(\varepsilon, 0)$ -SDP with probability $1 - \delta$ implies being (ε, δ) -SDP but not the other way around.⁷ Approximate SDP is, however, equivalent to the following definition.

⁷ See, for instance [12] or <https://differentialprivacy.org/flavoursofdelta/>.

Definition 13 (Smoothed SDP [46,11]). We say that a mechanism $\mathcal{M}$ is smoothed (ε, δ) -SDP if for every pair D, D' of adjacent databases, there are events $E = E(\mathcal{M}(D)), E' = E'(\mathcal{M}(D'))$ such that:

1. $\mathbb{P}(E), \mathbb{P}(E') \geq 1 - \delta$.
2. $\mathcal{M}(D)|E$ and $\mathcal{M}(D')|E'$ are mutually $e^{-\varepsilon}$ -dense.

Lemma 1 ([46,11]). A mechanism is (ε, δ) -SDP if and only if it is smoothed (ε, δ) -SDP.

We may consider $\mathcal{M}$ conditioned on the respective events as a new mechanism, but should note that the events are database-dependent, meaning that smoothed SDP is more akin to $\text{SIM}_{\forall\exists}$ -CDP than to SIM-CDP. If one sees the conditioned mechanism as a simulator $\mathcal{S}$ in the sense of $\text{SIM}_{\forall\exists}$ -CDP, then at first glance it might seem like smooth SDP is the same as requiring that the statistical distance between the mechanism and $\mathcal{S}$ is upper bounded by δ . This is however not the case, a fact that we rely upon in the next section. To ease notation, we introduce the following intermediate definition.

Definition 14 ($\text{SD}_{\forall\exists}$ -SDP). A mechanism $\mathcal{M}$ is $(\varepsilon, \delta, \hat{\delta})$ - $\text{SD}_{\forall\exists}$ -SDP if for every pair D, D' of adjacent databases, there are distributions $\mathcal{S}(D), \mathcal{S}(D')$ such that:

1. $SD(\mathcal{M}(D), \mathcal{S}(D)) \leq \delta$ and $SD(\mathcal{M}(D'), \mathcal{S}(D')) \leq \hat{\delta}$.
2. $\mathcal{S}(D)$ and $\mathcal{S}(D')$ are mutually $e^{-\varepsilon}$ -dense.

If $\delta = \hat{\delta}$, then we say that the mechanism is (ε, δ) - $\text{SD}_{\forall\exists}$ -SDP

Proposition 3. If a mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is (ε, δ) -SDP then it is also (ε, δ) - $\text{SD}_{\forall\exists}$ -SDP.

The proof of Proposition 3 is deferred to Appendix A.1.

Lemma 2. The converse of Proposition 3 does not hold. In particular, for all values ε, δ , (ε, δ) - $\text{SD}_{\forall\exists}$ -SDP directly implies $(\varepsilon, (e^\varepsilon + 1)\delta)$ -SDP and these parameters are tight. That is, there exists $\varepsilon, \delta, \mathcal{M}$ such that $\mathcal{M}$ is (ε, δ) - $\text{SD}_{\forall\exists}$ -SDP but not (ε, δ') -SDP for any $\delta' < (e^\varepsilon + 1)\delta$. More generally, $(\varepsilon, \delta, \hat{\delta})$ - $\text{SD}_{\forall\exists}$ -SDP directly implies $(\varepsilon, e^\varepsilon \max(\delta, \hat{\delta}) + \min(\delta, \hat{\delta}))$ -SDP and these parameters are tight in the same sense.

In other words, Lemma 2 says that (ε, δ) - $\text{SD}_{\forall\exists}$ -SDP does not directly imply (ε, δ) -SDP and that the δ -gap of this separation is $e^\varepsilon \delta$.

Proof. Let $\mathcal{M}$ be a mechanism which is $(\varepsilon, \delta, \hat{\delta})$ - $\text{SD}_{\forall\exists}$ -SDP for arbitrary parameters. This implies that, for every $S \subset \mathcal{R}$:

$$\begin{aligned} \mathbb{P}(\mathcal{M}(D) \in S) &\leq \mathbb{P}(\mathcal{S}(D) \in S) + \delta \\ &\leq e^\varepsilon \mathbb{P}(\mathcal{S}(D') \in S) + \delta \\ &\leq e^\varepsilon (\mathbb{P}(\mathcal{M}(D') \in S) + \hat{\delta}) + \delta \\ &= e^\varepsilon \mathbb{P}(\mathcal{M}(D') \in S) + e^\varepsilon \hat{\delta} + \delta. \end{aligned}$$

Similarly, in the other direction, we get $\mathbb{P}(\mathcal{M}(D') \in S) \leq \mathbb{P}(\mathcal{M}(D) \in S) + e^\varepsilon \delta + \hat{\delta}$. This shows the direct implication. Note that the implication indeed does hold for all $\varepsilon, \delta, \hat{\delta}$.

The tightness comes from showing a mechanism which (for some specific choice of parameters ε, δ) matches the bound. For simplicity, we show this only for the case for $\hat{\delta} = \delta$ but the argument is analogous for the other cases. Consider the binary case and let $\mathcal{S}$ be the randomized response mechanism [47]. That is, $\mathcal{R} = \mathcal{D} = \{0, 1\}$ and $\mathbb{P}(\mathcal{S}(D) = D) = \frac{e^\varepsilon}{e^\varepsilon + 1}$. Let $\mathcal{M}$ be the same mechanism but which outputs its input with δ extra probability. This assumes that $\frac{e^\varepsilon}{e^\varepsilon + 1} + \delta \leq 1$ but this is fine since the tightness only requires that there exists *some* parameters where the bound is tight. That is, set $\mathcal{M}$ such that $\mathbb{P}(\mathcal{M}(D) = D) = \mathbb{P}(\mathcal{S}(D) = D) + \delta = \frac{e^\varepsilon}{1 + e^\varepsilon} + \delta$. It is clear that $\mathcal{M}(D), \mathcal{M}(D'), \mathcal{S}(D), \mathcal{S}(D')$ are indeed distributions and that $SD(\mathcal{M}(D), \mathcal{S}(D)) \leq \delta$, and similarly for D' . Thus, for any two adjacent datasets $D, D' \in \{0, 1\}$, the same set of inequalities as above hold. That is, for every $S \subset \{0, 1\}$:

$$\mathbb{P}(\mathcal{M}(D) \in S) \leq e^\varepsilon \mathbb{P}(\mathcal{M}(D') \in S) + (e^\varepsilon + 1)\delta. \quad (5)$$

Therefore, we have confirmed that $\mathcal{M}$ is $(\varepsilon, (e^\varepsilon + 1)\delta)$ -SDP (by symmetry). To see that $\mathcal{M}$ is not (ε, δ') -SDP for any $\delta' < (e^\varepsilon + 1)\delta$, note that for any $D \in \{0, 1\}$ with $D' = 1 - D$:

$$\mathbb{P}(\mathcal{M}(D) = D) = \frac{e^\varepsilon}{1 + e^\varepsilon} + \delta, \quad (6)$$

and simultaneously, from the law of total probability,

$$\mathbb{P}(\mathcal{M}(D') = D) = \frac{1}{1 + e^\varepsilon} - \delta. \quad (7)$$

Therefore, by setting $D = 1, D' = 0, S = \{1\}$, we get

$$\begin{aligned} \mathbb{P}(\mathcal{M}(D) = D) &= \mathbb{P}(\mathcal{M}(1) = 1) = \frac{e^\varepsilon}{1 + e^\varepsilon} + \delta \\ &= e^\varepsilon \frac{1}{1 + e^\varepsilon} + \delta \\ &= e^\varepsilon (\mathbb{P}(\mathcal{M}(0) = 1) + \delta) + \delta \\ &= e^\varepsilon \mathbb{P}(\mathcal{M}(D') = D) + (e^\varepsilon + 1)\delta. \end{aligned}$$

Thus we have found a pair of adjacent D, D' and a subset $S \subset \mathcal{R}$ for which the inequality defining approximate SDP is equality and therefore the specified mechanism can not be (ε, δ') -SDP with any $\delta' < (e^\varepsilon + 1)\delta$. $\square$

A direct consequence of the mechanism in the proof above is that (ε, δ) -SD $_{\forall \exists}$ -SDP is strictly more expressive than (ε, δ) -SDP. This can be seen by setting the utility function as $u(\mathcal{M}(D), D) := \mathbb{1}\{\mathcal{M}(D) = D\}$. The utility gap of this

separation is $e^\varepsilon \delta$, meaning that there exists a utility function and choice of parameters where some (ε, δ) -SD $_{\forall\exists}$ -SDP mechanism has $e^\varepsilon \delta$ better utility than the best possible (ε, δ) -SDP mechanism.

Corollary 1. *For general $\varepsilon, \delta > 0$, (ε, δ) -SD $_{\forall\exists}$ -SDP is more expressive than $(\varepsilon, (e^\varepsilon + 1)\delta)$ -SDP. Further, (ε, δ) -SD $_{\forall\exists}$ -SDP is strictly more expressive than (ε, δ') -SDP for all $\delta' < (e^\varepsilon + 1)\delta$.*

5.2 The Implication from SIM $_{\forall\exists}$ -CDP to IND-CDP

The proof in MPRV that pure SIM $_{\forall\exists}$ -CDP directly implies IND-CDP is due to a simple expansion (the same as in the proof of Lemma 2) of the inequality in the SIM $_{\forall\exists}$ -CDP definition. Below we show that this same method for CapDP instead gives a separation with respect to expressiveness, due to the tightness in Lemma 2. This result is essentially the computational analog to Lemma 2, with computational distances instead of statistical ones. The proof of Proposition 4 is found in Appendix A.2.⁸

Proposition 4 (Direct implication from (ε, δ) -SIM $_{\forall\exists}$ -CDP to (ε, δ) -IND-CDP). *For every $\varepsilon, \delta \geq 0$, (ε, δ) -SIM $_{\forall\exists}$ -CDP directly implies $(\varepsilon, \delta + e^\varepsilon \text{negl})$ -IND-CDP for some negligible function negl . Furthermore, this implication is tight in the sense that (ε, δ) -SIM $_{\forall\exists}$ -CDP does not directly imply (ε, δ') -IND-CDP for any value of δ' noticeably smaller than $\delta + e^\varepsilon \text{negl}$.*

In the proposition above, and similar cases later, the negligible function negl included in the resolution is precisely the one which is guaranteed by the mechanism being the specific type of CDP. It is important that this is the case, since the CDP definitions require there exists *some* negligible function, not that any negligible function may be used for any mechanism fulfilling the definition. Therefore, we think of the negl as the specific function derived when proving that $\mathcal{M}$ is SIM $_{\forall\exists}$ -CDP. Since this function is fixed, the choice of ε decides whether the $e^\varepsilon \text{negl}$ -term itself is negligible or not. Indeed, we know that if $e^\varepsilon \text{negl}$ is non-negligible then ε is superlogarithmic, but not every superlogarithmic ε results in a non-negligible $e^\varepsilon \text{negl}$ -term. In the extreme case, one could have $\varepsilon(\kappa) = \kappa$ and $\text{negl}(\kappa) = 1/2^{2^\kappa}$, which makes $e^\varepsilon \text{negl}$ negligible.

Note that the mechanisms in the proof ($\mathcal{M}$ and $\mathcal{S}$) might not be efficiently computable (it depends on parameter choices) but this has no effect on the proof, since neither CDP definition demands efficient mechanisms. Further, there are parameter choices such that the mechanisms are efficient and still yield the separation. Precisely as for SDP and SD $_{\forall\exists}$ -SDP, the construction in the proof above yields a separation in terms of expressiveness as well.

Corollary 2. *For all $\delta < 1$, (ε, δ) -SIM $_{\forall\exists}$ -CDP directly implies (ε, δ) -IND-CDP if and only if $\varepsilon \in O(\log(\kappa))$. Further, for $\varepsilon \in \omega(\log(\kappa))$ (ε, δ) -SIM $_{\forall\exists}$ -CDP*

⁸ For the special case of logarithmic ε and arbitrary δ , the direct implication (without tightness) was proven already in [38].

is strictly more expressive than (ε, δ) -IND-CDP. If $\varepsilon \in \omega(\log(\kappa))$ then (ε, δ) -SIM $_{\forall\exists}$ -CDP directly implies $(\varepsilon, \delta + \hat{\delta})$ -IND-CDP, where $\hat{\delta}$ is non-negligible.

Remark 2. The separation in the special case when $\delta = 0$ can also be proven by the following simple argument, which was shown to us by an anonymous reviewer at Crypto'25. Consider a superlogarithmic ε and the mechanism $\mathcal{M}$ which simply outputs the input database. The simulator $\mathcal{S}$ which outputs D with probability $1 - e^{-\varepsilon}$ and otherwise outputs D' is indistinguishable from $\mathcal{M}$, since $e^{-\varepsilon}$ is negligible. At the same time, $\mathcal{S}(D)$ and $\mathcal{S}(D')$ are mutually $e^{-\varepsilon}$ -dense: $\mathbb{P}(\mathcal{S}(D) = D) = 1 - e^{-\varepsilon} \leq 1 = e^{\varepsilon} \mathbb{P}(\mathcal{S}(D') = D)$. Thus $\mathcal{M}$ is $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP but clearly not $(\varepsilon', 0)$ -IND-CDP for any ε' . This tells us that SIM $_{\forall\exists}$ -CDP gives us no guarantees at all when ε is superlogarithmic, whereas IND-CDP might still give some guarantees (though very weak ones).

5.3 The Implication from IND-CDP to SIM $_{\forall\exists}$ -CDP

We start by reviewing MPRV's result for CpuDP by recalling the dense model theorem (DMT) and then treat extending their method to CapDP. The DMT is a reduction from having a dense model to pseudodensity and in this reduction there is a difference in the collections of tests, necessitating a notion of relative complexity between the test collections.

Definition 15 (Relative complexity [34,42]). *For a collection $\mathcal{T}$ of functions, a function h and $C \in \mathbb{N}$, we say that h has complexity C relative to $\mathcal{T}$ if there are functions $T_1, \dots, T_k \in \mathcal{T}$ (not necessarily distinct) with $k \leq C$ and h can be defined by combining $T_1, \dots, T_k$ using at most C of the following operations: sum, product, multiplication by constant, and application of a boolean threshold function.*

Below is the dense model theorem, based on the formulation in [33]. For a proof the theorem, we refer to [43]. Also note that the theorem is stated and proven with respect to the uniform distribution (pseudodensity in the uniform distribution implies the existence of a model that is truly dense in the uniform distribution) but that the proof of [43] also applies when the uniform distribution is replaced by any other. This fact is used in MPRV [41].

Theorem 1 (Dense model theorem (DMT)). *For a collection $\mathcal{T}$ of tests on a finite universe $\mathcal{R}$, let $\mathcal{T}^k$ denote the class of all tests with complexity k relative to $\mathcal{T}$. For every collection $\mathcal{T}$ and $\alpha, \beta > 0$, there exists a $k \in O(\frac{1}{\alpha^2} \log(\frac{1}{\beta}))$ such that if X is a distribution on $\mathcal{R}$ which is $(\beta, \alpha\beta)$ -pseudodense in the uniform distribution with respect to $\mathcal{T}^k$ then X has a β -dense α -model in the uniform distribution on $\mathcal{R}$ with respect to $\mathcal{T}$.*

For the application of the DMT to CpuDP, one would naturally want to set $\beta = e^{-\varepsilon\kappa}$, $\alpha = \text{negl}(\kappa)$ and have both $\mathcal{T}$ and $\mathcal{T}^k$ be $\mathcal{T}_{\text{poly}}$, the set of tests computable by non-uniform polynomial-time algorithms. This, however, will not work right away, since α being negligible makes k superpolynomial, meaning

that $\mathcal{T}$ and $\mathcal{T}^k$ cannot both be $\mathcal{T}_{poly}$. Therefore, one must first make use of the fact that negligible indistinguishability (and pseudodensity) with respect $\mathcal{T}_{poly}$ implies that there is some superpolynomial function s such that negligible indistinguishability also holds against all non-uniform algorithms of runtime at most s . The same holds also for distinguishability bounds of the form $\delta + \text{negl}$. This equivalence between indistinguishability with respect *arbitrary* polynomial time tests and with respect to tests of some *fixed* superpolynomial runtime bound stems from arguments in Bellare’s note about negligible functions [7].⁹ In the end, we have that $(\varepsilon, 0)$ -IND-CDP is equivalent to there existing some superpolynomial s such that $\mathcal{M}(D)$ and $\mathcal{M}(D')$ are mutually $(e^{-\varepsilon}, 1/s(\kappa))$ -pseudodense with respect to $\mathcal{T}_{s(\kappa)}$. At the same time, $\mathcal{M}$ being $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP directly follows if one can show that the SIM $_{\forall\exists}$ -CDP definition holds with respect all tests of some fixed superpolynomial complexity. Therefore, a superpolynomial relative complexity in the DMT is not prohibitive to its use in relating IND-CDP and SIM $_{\forall\exists}$ -CDP, as showed in MPRV, as long as $\mathcal{T}$ and $\mathcal{T}^\kappa$ both consists of all non-uniform tests of complexity below some (different) fixed superpolynomial.

When applying the DMT-based proof of MPRV to CapDP then it leads to IND-CDP implying the existence of two distributions $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$ such that they are mutually $e^{-\varepsilon_\kappa}$ -pseudodense and simultaneously $CD(\mathcal{M}_\kappa(D), \mathcal{S}_\kappa(D)) \leq e^{\varepsilon_\kappa}(\delta_\kappa + \text{negl}(\kappa))$ and $CD(\mathcal{M}(D'), \mathcal{S}(D')) \leq e^{\varepsilon_\kappa}\delta_\kappa + e^{3\varepsilon_\kappa}(\delta_\kappa + \text{negl}(\kappa))$. The existence of such $\mathcal{S}$ does not in itself directly say anything about $\mathcal{M}$ being SIM $_{\forall\exists}$ -CDP. However, if one is always able to translate all of the computational distances to statistical distances by introducing a negligible slack in computational distance (made precise below) then $\mathcal{M}$ would be computationally close to a pair of distributions that are statistically close to being pure SDP. Then by the tight direct implication from SD $_{\forall\exists}$ -SDP to SDP from above, one arrives at $\mathcal{M}$ being SIM $_{\forall\exists}$ -CDP, with increased parameters. That a general ‘CD-to-SD’ translation exists in the way we need it follows as a corollary of the Hybrid Indistinguishability Lemma by Bitansky and Geier [9]. Using this, we prove that the proof technique of MPRV does indeed give a direct implication from IND-CDP to SIM $_{\forall\exists}$ -CDP also for general δ , but with a significant (although non-trivial) δ -gap.

One should remark that we merely derive a upper bound on the parameter gap resulting from using precisely the proof strategy in MPRV. It is conceivable that by using a different one, it is possible to derive a tighter direct implication from IND-CDP to SIM $_{\forall\exists}$ -CDP. That is, we leave it as an open question whether

⁹ More details on using Bellare’s results with respect to adversarial boundedness can be found in [1] (proof of their Theorem 69). The outline is that Bellare shows that for all countable collections of negligible functions, there exist some other negligible function which dominates them. Then one need only note that $\mathcal{T}_{poly}$ behaves in a sense as if it is countable and show that the negligible function dominating these tests, with the negligible function associated to each test being the distinguishing advantage, cannot be the advantage of any test in $\mathcal{T}_{poly}$.

our δ -gap is optimal or if there exists a tighter direct implication to be found. In particular, it is left open whether there needs to be a δ -gap at all (i.e. if also in CapDP, $\text{SIM}_{\forall\exists}$ -CDP and IND-CDP are equivalent when $\varepsilon \in O(\log(\kappa))$). Now we formalize the results outlined above. The first step of the DMT-based proof is to use the lemma below. The proof of the lemma is found in Section 5.4, and we note that it is essentially unchanged from the first part of the proof of Theorem 3 in MPRV [41].

Lemma 3 (Generalization of partial Theorem 3 in MPRV [41]). *For every $\varepsilon, \delta \geq 0$, if a mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is (ε, δ) -IND-CDP then for every sufficiently large κ and all pairs D, D' of adjacent datasets in $\mathcal{D}_\kappa$ there exists distributions $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$ over $\mathcal{R}_\kappa$ such that:*

1. $\mathcal{S}_\kappa(D)$ and $\mathcal{S}_\kappa(D')$ are mutually $e^{-\varepsilon\kappa}$ -dense, and,
2. $CD(\mathcal{M}_\kappa(D), \mathcal{S}_\kappa(D)) \leq e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$, and,
3. $CD(\mathcal{M}_\kappa(D'), \mathcal{S}_\kappa(D')) \leq e^{\varepsilon\kappa}\delta_\kappa + e^{3\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$.

Here negl is the specific negligible function arising from $\mathcal{M}$ being IND-CDP.

Lemma 4 (Hybrid indistinguishability lemma [9]). *For probability distributions X, Y on sample spaces $\mathcal{X}, \mathcal{Y}$, respectively, and parameters $\delta, \eta \in (0, 1), s \in \mathbb{N}$, if $CD_{\mathcal{T}_s}(X, Y) \leq \delta$ then there exists a third distribution Z such that $SD(Z, Y) \leq \delta$ and $CD_{\mathcal{T}_{s'}}(X, Z) \leq \eta$, with*

$$s' = \frac{s(\eta/(1-\delta))^2}{128 \cdot (\log(|\mathcal{X}|) + \log(|\mathcal{Y}|) + 1)}.$$

In our setting, we have s being superpolynomial in κ and therefore we need only ascertain that there exists a negligible η such that $s'(\kappa)$ is also superpolynomial. Since we assume that X, Y, Z are on a sample space $\mathcal{R}$ which has polynomially large elements, we get $\mathcal{X} = \mathcal{Y} = \mathcal{R}$ with $\log(|\mathcal{R}|)$ being polynomial. This in turn makes the denominator of the formula for s' polynomial and we denote this polynomial by $p(\kappa)$. Now note that for every superpolynomial s there exists a negligible η such that $s' = \frac{s \cdot (\eta/(1-\delta))^2}{p(\kappa)}$ is also superpolynomial.¹⁰ This gives us the corollary below, as a special case of Lemma 4.

Corollary 3. *For every two probability distributions X, Y on $\mathcal{R}$ and $\delta \in (0, 1)$, if $CD(X, Y) \leq \delta$ then there exists a third distribution Z on $\mathcal{R}$ and negligible function negl such that $SD(Z, Y) \leq \delta$ and $CD(X, Z) \leq \text{negl}$.*

We are now ready to present the theorem showing a direct implication from $\text{SIM}_{\forall\exists}$ -CDP to IND-CDP for CapDP.

Theorem 2. *For every $\varepsilon, \delta \geq 0$, a mechanism $\mathcal{M}$ being (ε, δ) -IND-CDP directly implies that it is also (ε, δ') - $\text{SIM}_{\forall\exists}$ -CDP, with $\delta' = e^{2\varepsilon}\delta + e^{4\varepsilon}(\delta + \text{negl}) + e^\varepsilon(\delta + \text{negl})$.*

¹⁰ For instance, we can set η to $\frac{(1-\delta)\sqrt{p}}{s^{1/3}}$, which makes $s' = s^{1/3}$, which clearly is superpolynomial.

Proof. Let $\mathcal{M}$ be an (ε, δ) -IND-CDP mechanism ensemble. We apply Lemma 3 and get $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$. For large enough κ , we have that $CD(\mathcal{M}_\kappa(D), \mathcal{S}_\kappa(D)) \leq e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$ and $CD(\mathcal{M}_\kappa(D'), \mathcal{S}_\kappa(D')) \leq e^{\varepsilon\kappa}\delta_\kappa + e^{3\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$. By Corollary 3, we get the existence of two new distributions $\mathcal{S}_\kappa^o(D), \mathcal{S}_\kappa^o(D')$ such that:

$$\begin{aligned} & - CD(\mathcal{M}_\kappa(D), \mathcal{S}_\kappa^o(D)) \leq \text{negl}'(\kappa) \text{ and } CD(\mathcal{M}_\kappa(D'), \mathcal{S}_\kappa^o(D')) \leq \text{negl}'(\kappa), \text{ and,} \\ & - SD(\mathcal{S}_\kappa(D), \mathcal{S}_\kappa^o(D)) \leq e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa)) \text{ and } SD(\mathcal{S}_\kappa(D'), \mathcal{S}_\kappa^o(D')) \leq e^{\varepsilon\kappa}\delta_\kappa + e^{3\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa)). \end{aligned}$$

Note that negl' is a potentially different negligible function than negl . Since $\mathcal{S}^o$ is $\text{SD}_{\forall\exists}$ -SDP, by the general case in Lemma 2 we get that $\mathcal{S}_\kappa^o(D)$ and $\mathcal{S}_\kappa^o(D')$ are mutually $(\varepsilon_\kappa, e^{\varepsilon\kappa}(e^{\varepsilon\kappa}\delta_\kappa + e^{3\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))) + e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa)))$ -dense. Thus $\mathcal{M}$ is $(\varepsilon, e^{2\varepsilon}\delta + e^{4\varepsilon}(\delta + \text{negl}) + e^\varepsilon(\delta + \text{negl}))$ - $\text{SIM}_{\forall\exists}$ -CDP. $\square$

For the application of Lemma 2, it is important to note that whilst that lemma is phrased and proved for mechanisms, its proof consists solely of showing that two distributions each having a fixed statistical distance α to one of two other distributions that are mutually β -dense is a stricter requirement on the original distributions than them being (β, α) -dense. It is precisely this observation which is used in the proof of Theorem 2. The notation $\mathcal{S}^o$ is motivated by seeing them as ‘outer’ simulators, as opposed to $\mathcal{S}$ being ‘inner’ simulators, in the sense of being further removed from $\mathcal{M}$, which then represents the ‘outside’ reality. As expected, if $\delta = 0, \varepsilon \in O(\log(\kappa))$ then Corollary 3 is not needed and Theorem 2 recovers the corresponding direct implication in MPRV (their Theorem 3).

5.4 Proof of Lemma 3

We start by stating a reformulation (using the terminology of [24]) of the main lemma used in the proof (MPRV’s Theorem 2), which is an extension of the dense model theorem (DMT). We refer to this lemma as the extended DMT.

Lemma 5 (Reformulation of Theorem 2 in MPRV [41]). *Let X_κ, Y_κ be distributions over a finite sample space $\mathcal{R}_\kappa$, $\varepsilon_\kappa \geq 0, \delta_\kappa \in (0, 1]$ be constants, $\mathcal{T}$ be an arbitrary collection of tests over $\mathcal{R}_\kappa$ and $\mathcal{T}^k$ be the collection of tests with complexity $k = O(\frac{1}{\delta_\kappa^2} \log(\frac{e^{\varepsilon_\kappa}}{\delta_\kappa}))$ relative to $\mathcal{T}$. It holds that:*

1. DMT: *If X_κ and Y_κ are mutually $(e^{-\varepsilon_\kappa}, \delta_\kappa)$ -pseudodense with respect to $\mathcal{T}^k$ then there exists a third distribution Z_κ (the model of X_κ in Y_κ) over $\mathcal{R}_\kappa$ such that Z_κ is $e^{-\varepsilon_\kappa}$ -dense in Y_κ and $CD_{\mathcal{T}}(X_\kappa, Z_\kappa) \leq e^{\varepsilon_\kappa}\delta_\kappa$.*
2. Extension: *If additionally, Y_κ is $e^{-\varepsilon_\kappa}$ -dense in X_κ then it can also be guaranteed that Y_κ is $e^{-\varepsilon_\kappa}$ -dense in Z_κ .*

In Lemma 5, the DMT translates a distinguishing-parameter of δ_κ in the pseudodensity to a distinguishing-parameter of $e^{\varepsilon_\kappa}\delta_\kappa$ in the model. It is unavoidable to have an increase but the factor varies between different works (for

instance, in MPRV the factor is fixed as 4). It is however known that having the factor $e^{\varepsilon\kappa}$ is optimal in the sense that the DMT becomes false for all smaller factors [49,48,33], which is why we use it. Note that if one uses the factor 4, as in MPRV, the direct implication we derive remains qualitatively the same, although with a different δ -gap. In particular, the δ -gap remains a polynomial of $e^\varepsilon\delta$ but with smaller exponents and larger constants.

Proof (Lemma 3 [41]). For a positive function s , $\mathcal{T}_{s(\kappa)}$ denotes the set of all tests computable by some non-uniform algorithm with runtime at most $s(\kappa)$. That $\mathcal{M}$ is (ε, δ) -IND-CDP implies that there exists a superpolynomial function s in κ such that for all adjacent D, D' and large enough κ , $\mathcal{M}_\kappa(D)$ and $\mathcal{M}_\kappa(D')$ are mutually $(e^{-\varepsilon\kappa}, \delta_\kappa + \text{negl}(\kappa))$ -pseudodense with respect to $\mathcal{T}_{s(\kappa)}$. Therefore we can apply the DMT part of Lemma 5 which gives the existence of a distribution $\mathcal{S}_\kappa(D)$ which is $e^{-\varepsilon\kappa}$ -dense in $\mathcal{M}_\kappa(D')$ and has $CD_{\mathcal{T}_{s'(\kappa)}}(\mathcal{M}_\kappa(D), \mathcal{S}_\kappa(D)) \leq e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$ for some $s'(\kappa) \leq s(\kappa) \cdot \Omega(\delta_\kappa^2 / \log(e^{\varepsilon\kappa}/\delta_\kappa))$. Note that s' is also superpolynomial. This gives us for every $T \in \mathcal{T}_{s'(\kappa)}$:

$$\begin{aligned} \mathbb{P}(T(\mathcal{M}_\kappa(D')) = 1) &\leq e^{\varepsilon\kappa} \mathbb{P}(T(\mathcal{M}_\kappa(D)) = 1) + \delta_\kappa + \text{negl}(\kappa) \\ &\leq e^{\varepsilon\kappa} (\mathbb{P}(T(\mathcal{S}_\kappa(D)) = 1) + e^{\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))) + \delta_\kappa + \text{negl}(\kappa) \\ &= e^{\varepsilon\kappa} (\mathbb{P}(T(\mathcal{S}_\kappa(D)) = 1) + e^{2\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa)) + \delta_\kappa). \end{aligned}$$

Now $\mathcal{M}_\kappa(D')$ is $(e^{-\varepsilon\kappa}, \delta_\kappa + e^{2\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa)))$ -pseudodense in $\mathcal{S}_\kappa(D)$, which together with $\mathcal{S}_\kappa(D)$ being $e^{-\varepsilon\kappa}$ -dense in $\mathcal{M}_\kappa(D')$ means that we can apply the extension part of Lemma 5. Therefore there exists a distribution $\mathcal{S}_\kappa(D')$ which is mutually $e^{-\varepsilon\kappa}$ -dense in $\mathcal{S}_\kappa(D)$ and $CD_{\mathcal{T}_{s''(\kappa)}}(\mathcal{M}_\kappa(D'), \mathcal{S}_\kappa(D')) \leq e^{\varepsilon\kappa} \cdot (\delta_\kappa + e^{2\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))) = e^{\varepsilon\kappa}\delta_\kappa + e^{3\varepsilon\kappa}(\delta_\kappa + \text{negl}(\kappa))$, again with $s''(\kappa) \leq s'(\kappa) \cdot \Omega(\delta_\kappa^2 / \log(e^{\varepsilon\kappa}/\delta_\kappa))$. Now since s, s', s'' are all superpolynomial, the lemma follows. $\square$

6 Extensions of Barrier Results

At TCC in 2011, Groce, Katz and Yerukhimovich [29] establish two barriers to expressiveness separations between $(\varepsilon, \text{negl})$ -SDP and CpuDP:

- *‘Exotic-crypto’ barrier:* Only assuming fully black-box access to a one-way function is not enough to separate $(\varepsilon, \text{negl})$ -SDP and CpuDP (in terms of expressiveness) in the central model (with similar efficiency), for any (even inefficient) utility function. This means that in order to find a separation, one must use more “exotic cryptography” [26].
- *L_p -utility barrier:* Under any complexity assumption, $(\varepsilon, \text{negl})$ -SDP and CpuDP cannot be separated for efficient (PPT) mechanisms by using a utility function which is a bound on the v -th power of an L_p -distance, when the size of the mechanism range is independent of κ .

These barriers are the reason that [10,26] use contrived tasks and rely on stronger complexity assumptions than the black-box use of one-way functions. It is worth noting that whilst the results by [29] are called *barriers* in the literature, they are in fact expressiveness implications from IND-CDP to SDP. That is, they show that for two combinations of a class of complexity assumptions and a class of utility functions (computational tasks), if a task from that class can be solved with IND-CDP under assumptions from the class of assumptions, then it can also be solved in SDP. The reason these results are referred to as barriers is that the search for practically relevant separations consists of precisely finding tasks and complexity assumptions where this is not the case. We show the following two results, which are formalized in later subsections:

1. *Restriction of L_p -utility barrier proof:* For $(\varepsilon, 0)$ -SDP and $(\varepsilon, 0)$ -SIM $_{\forall\exists}$ -CDP, [29]’s proof of the L_p -utility barrier requires that ε is logarithmic. This applies again (and independently) for the L_p -utility barrier translation from SIM $_{\forall\exists}$ -CDP to IND-CDP.
2. *Extension of both barriers to CapDP:* For (ε, δ) -SDP and (ε, δ) -IND-CDP, both barriers hold also for any value of δ . In the L_p -utility barrier, the proof from SIM $_{\forall\exists}$ -CDP to SDP requires logarithmic ε and in its extension to IND-CDP, the barrier inherits the same δ -gap as the best known direct implication from IND-CDP to SIM $_{\forall\exists}$ -CDP.

Note that the barriers do not contradict the separation we show earlier, since the barriers concern implications *from* IND-CDP *to* SDP (if a task can be solved with IND-CDP then it can be solved with SDP, and thus with SIM-CDP and SIM $_{\forall\exists}$ -CDP also) whereas our separation goes in the other direction (there are tasks that can be solved with SIM $_{\forall\exists}$ -CDP but not IND-CDP when ε is large).

Two remarks about our results are in order. Firstly, that the L_p -utility barrier proof does not allow arbitrarily large ε is likely of very limited importance to the power of the barrier, since separations between SDP and CDP in settings of superlogarithmic ε are hardly to be seen as relevant for practice. Secondly, that the barriers extend also to the setting of non-zero δ means that we cannot hope to circumvent the barriers simply by introducing a small, but potentially non-negligible, δ . Whereas a logarithmic bound on ε is seemingly a very small weakening of the power of the barriers, the applicability when $\delta > 0$ might be a significant strengthening.

6.1 The Exotic-crypto Barrier

As explained above, the exotic-crypto barrier says essentially that only black-box use of one-way functions (OWFs) is not enough to separate SDP and CDP. This is formalized by assuming that $\mathcal{M}$ is a fully black-box construction from OWFs, meaning it has only oracle access to some OWF $\mathcal{O}$, and then showing that this mechanism can in a sense be simulated by an SDP mechanism. When the mechanism $\mathcal{M}$ has oracle access to $\mathcal{O}$, we denote it by $\mathcal{M}^{\mathcal{O}}$, as is common.

We remark that $\mathcal{M}^\mathcal{O}$ is IND-CDP for every OWF $\mathcal{O}$ and we measure the utility of $\mathcal{M}$ under a random choice of $\mathcal{O}$ from the set of one-way permutations.

Theorem 3 (Extension of exotic-crypto barrier). *For any $\varepsilon, \delta \geq 0$, if there exists a fully black-box construction $\mathcal{M}$ of an (ε, δ) -IND-CDP mechanism from one-way functions, then there exists two negligible functions $\text{negl}, \text{negl}'$ and an $(\varepsilon, \delta + \text{negl})$ -SDP mechanism $\mathcal{M}'$ such that for all $D \in \mathcal{D}$ and (real-valued, potentially inefficient) utility functions $u : \mathcal{D} \times \mathcal{R} \rightarrow \mathbb{R}$:*

$$|\mathbb{E}[u(D, \mathcal{M}^\mathcal{O}(D))] - \mathbb{E}[u(D, \mathcal{M}'_\kappa(D))]| \leq \text{negl}'(\kappa),$$

The first expectation is taken over the mechanism's random coins and the random choice of the one-way function $\mathcal{O}$, and the right expectation is taken over only the mechanism's coins.

The proof of the theorem is exactly the same as that of the same barrier in [29], except for extra details in the few steps where the negligible functions are derived and when it is used that $\mathcal{M}$ is IND-CDP. For this reason, we defer the proof of Theorem 3 to Appendix A.3.

6.2 The L_p -utility Barrier

The L_p -utility barrier applies to a larger set of utilities than just bounded L_p norms, namely the *average* (p, v) -error, denoted $\sigma_{p,v}(q, D, \mathcal{M}_\kappa)$. The average error is instantiated with respect to some query function $q : \mathcal{D} \rightarrow \mathcal{R}$, representing the ‘correct’ output the mechanism is supposed to give on the specific database D . For $p > 0, v \geq 1$, $\sigma_{p,v}$ is the expectation of the v 'th power of the L_p -distance between $q(D)$ and $\mathcal{M}_\kappa(D)$.¹¹ The average error can be turned into a utility function u by having u output 1 iff the average error is below some specified threshold. Such utility functions allow capturing, for instance, the mean squared error. For the rest of this section, we fix $\mathcal{R}$ as $\mathbb{R}^n$ with $n \in \mathbb{N}^+$ independent of κ . Apart from the choice of $\mathcal{R}$ and type of utility function, there are two more restrictions in the L_p -utility barrier. Firstly, the CDP mechanism is assumed to be *efficient* (PPT) and secondly, it must have an average (p, v) -error that is polynomially bounded.

Theorem 4 (Extension of the L_p -utility barrier). *For any fixed $p > 0, v \geq 1$ and any $\varepsilon \in O(\log(\kappa)), \delta \geq 0$, let $\mathcal{M} = \mathcal{M}_\kappa : \mathcal{D} \rightarrow \mathbb{R}^n$ be an efficient (ε, δ) -SIM $_{\forall\exists}$ -CDP mechanism with $\sigma_{p,v}(q, D, \mathcal{M}_\kappa) \leq \text{err}(\kappa)$ for all $D \in \mathcal{D}, \kappa \in \mathbb{N}$, with err being some polynomial. Then there exists an efficient $(\varepsilon, \delta + \text{negl})$ -SDP mechanism $\mathcal{M}'$ with $\sigma_{p,v}(q, D, \mathcal{M}'_\kappa) \leq \text{err}(\kappa) + \text{negl}'(\kappa)$ for all $D \in \mathcal{D}, \kappa \in \mathbb{N}$.*

In [29], the theorem is stated for $\delta = 0$ and with SIM-CDP instead of SIM $_{\forall\exists}$ -CDP. As is noted by the authors, however, the proof only analyzes adjacent database pairs independently and thus the order of quantifiers does not matter, therefore the theorem applies equally well to SIM $_{\forall\exists}$ -CDP. Of course,

¹¹ In other words, it is the v 'th moment of the L_p -error.

by the lossless direct implication from SIM-CDP to $\text{SIM}_{\forall\exists}$ -CDP, the theorem when stated for $\text{SIM}_{\forall\exists}$ -CDP immediately applies also to SIM-CDP. Stating the theorem for $\text{SIM}_{\forall\exists}$ -CDP is however instructive when it comes to treating IND-CDP, since in [29] the extension of the barrier to IND-CDP goes via the direct implication from IND-CDP to $\text{SIM}_{\forall\exists}$ -CDP.

Just as for the other barrier, the proof carries over essentially unchanged to the approximate case. However, since the proof of the L_p -utility barrier is somewhat long and technical and only one single step in the proof involves the parameters ε, δ , we give a proof sketch rather than a full proof of the theorem above. In particular, every single step of the proof of our theorem above is identical to that in [29]’s proof of the L_p -utility barrier, except for the one where one plugs in the mutual (α, β) -density as guaranteed by the $\text{SIM}_{\forall\exists}$ -CDP definition. Again, we defer the proof sketch to Appendix A.4, due to similarity to [29]’s proof.

We also note that for the proof of the L_p -utility barrier, it is necessary that ε is logarithmic and this is for the same reasons this restriction arise in earlier results of this work (one needs at some point in the proof that $e^\varepsilon \cdot \text{negl}$ is negligible for arbitrary negl). This restriction was not identified in [29]. When extending the barrier to be between SDP and IND-CDP by using a direct implication from IND-CDP to $\text{SIM}_{\forall\exists}$ -CDP (as is done in [29]), again this requirement arise. In the case of $\delta = 0$, the direct implication of MPRV [41] can be used. In the case of $\delta > 0$, the only known direct implication is the one we show earlier in this work, which has a significant δ -gap. This δ -gap is then inherited by the barrier.

7 Conclusions and Open Problems

The current state of knowledge is summarized in Table 1, with areas affected by our work in bold. Parameter regimes that are simply marked by ‘Open’ are entirely open in the sense that there is no result that relate the two definitions in question for such parameters. All results are listed in more detail in Table 2. The results for IND-CDP and $\text{SIM}_{\forall\exists}$ -CDP with non-negligible δ or superlogarithmic ε are from this work, the other results are from [41, 10, 26]. From our results arises one main open question, namely if the δ -gap in the direct implication from IND-CDP to $\text{SIM}_{\forall\exists}$ -CDP is optimal.

Open problem 1 *Does there, for arbitrary parameter choices, exist a direct implication from IND-CDP to $\text{SIM}_{\forall\exists}$ -CDP with a δ -gap below $e^{2\varepsilon}\delta + e^{4\varepsilon}(\delta + \text{negl}) + e^\varepsilon(\delta + \text{negl}) - \delta$?*

It is the openness of this problem that cause the divided elements in Table 1 since if the problem is resolved in the negative (or even positively with a tight non-zero δ -gap), then $\text{SIM}_{\forall\exists}$ -CDP and IND-CDP are incomparable, meaning that neither definition directly implies the other without δ -gap. If the problem is resolved in the positive then depending on the δ -gap of the resolution, the

Table 1. Table of known relations between the approximate CDP definitions in various parameter regimes. The results are stated for the case when both definitions use the same parameters and under arbitrary complexity assumptions. ‘Sep’ stands for separated, i.e. the first definition implies the other but not the other way around, ‘Eq’ for equivalent, ‘EE’ for equally expressive (up to negligible differences in utility), ‘ME’ for one being (strictly) more expressive than the other and ‘Inf’ stands for strictly more expressive for efficient mechanisms. ‘I’ denotes ‘incomparable’, meaning that neither definition implies the other.

SDP to SIM-CDP (by [41,10])								
	$\delta = 0$		$\delta = \text{negl}$		$\delta = 1/\text{poly}$		$\delta < 1$ constant	
	DI	Expr.	DI	Expr.	DI	Expr.	DI	Expr.
ε constant	Sep	EE & Inf	Sep	EE & Inf	Sep	EE & Inf	Sep	Open
$\varepsilon \in O(\log(\kappa))$	Sep	EE & Inf	Sep	EE & Inf	Sep	EE & Inf	Sep	Open
$\varepsilon \in \omega(\log(\kappa))$	Sep	Open	Sep	Open	Sep	Open	Sep	Open
SIM-CDP to SIM _∃ -CDP (by [26,41])								
	$\delta = 0$		$\delta = \text{negl}$		$\delta = 1/\text{poly}$		$\delta < 1$ constant	
	DI	Expr.	DI	Expr.	DI	Expr.	DI	Expr.
ε constant	Sep	ME	Sep	ME	Open	Open	Open	Open
$\varepsilon \in O(\log(\kappa))$	Open	Open	Open	Open	Open	Open	Open	Open
$\varepsilon \in \omega(\log(\kappa))$	Open	Open	Open	Open	Open	Open	Open	Open
SIM _∃ -CDP to IND-CDP (by [41] and this work)								
	$\delta = 0$		$\delta = \text{negl}$		$\delta = 1/\text{poly}$		$\delta < 1$ constant	
	DI	Expr.	DI	Expr.	DI	Expr.	DI	Expr.
ε constant	Eq	EE	Eq	EE	Sep/Eq	ME/EE	Sep/Eq	ME/EE
$\varepsilon \in O(\log(\kappa))$	Eq	EE	Eq	EE	Sep/Eq	ME/EE	Sep/Eq	ME/EE
$\varepsilon \in \omega(\log(\kappa))$	Sep	ME	Sep	ME	Sep/I	ME/I	Sep/I	ME/I

Table 2. Summary of implications and separations in central-model CDP, not including the barriers. For the new implication from IND-CDP to SIM_∃-CDP, $\delta^* = e^{2\varepsilon}\delta + e^{4\varepsilon}(\delta + \text{negl}) + e^\varepsilon(\delta + \text{negl})$. The results marked by ‘*’ are conditional on assumptions.

Result	Parameter restrictions	δ -gap
$(\varepsilon, \delta)\text{-SDP} \xRightarrow{(DI)} (\varepsilon, \delta)\text{-SIM-CDP}$	-	0
$(\varepsilon, \delta)\text{-SIM-CDP} \not\xRightarrow{(DI)} (\varepsilon, \delta)\text{-SDP}$	-	0
$(\varepsilon, 0)\text{-SIM-CDP} \not\xRightarrow{(ME)} (\varepsilon', 1/\text{poly})\text{-SDP}$ [10]*	$\varepsilon > 0, \varepsilon' \in O(\log(\kappa))$	$1/\text{poly}$
$(\varepsilon, \delta)\text{-SIM-CDP} \xRightarrow{(DI)} (\varepsilon, \delta)\text{-SIM}_{\exists}\text{-CDP}$	-	0
$(\varepsilon, 0)\text{-SIM}_{\exists}\text{-CDP} \not\xRightarrow{(ME)} (\varepsilon', \delta')\text{-SIM-CDP}$ [26]*	$\varepsilon, \varepsilon' > 0$ const., $\delta' \leq 1/\kappa^{27}$	$1/\kappa^{27}$
$(\varepsilon, 0)\text{-SIM}_{\exists}\text{-CDP} \xRightarrow{(DI)} (\varepsilon, 0)\text{-IND-CDP}$ [41]	$\varepsilon \in O(\log(\kappa))$ (Cor. 2)	-
$(\varepsilon, \delta)\text{-SIM}_{\exists}\text{-CDP} \xRightarrow{(DI)} (\varepsilon, \delta^*)\text{-IND-CDP}$ (Prop. 4)	$\delta' \geq \delta + e^\varepsilon \text{negl}$	$e^\varepsilon \text{negl}$
$(\varepsilon, 0)\text{-IND-CDP} \xRightarrow{(DI)} (\varepsilon, 0)\text{-SIM}_{\exists}\text{-CDP}$ [41]	$\varepsilon \in O(\log(\kappa))$	-
$(\varepsilon, \delta)\text{-IND-CDP} \xRightarrow{(DI)} (\varepsilon, \delta^*)\text{-SIM}_{\exists}\text{-CDP}$ (Thm. 2)	-	$\delta^* - \delta$
$(\varepsilon, \delta')\text{-IND-CDP} \not\xRightarrow{(ME)} (\varepsilon, \delta)\text{-SIM}_{\exists}\text{-CDP}$ (Prop. 4)	$\delta' < \delta + e^\varepsilon \text{negl}$	$e^\varepsilon \text{negl}$

definitions may either be separated, incomparable or equivalent, in the various parameter regimes. We remark that in order to resolve the open problem above, one would need to either make additional assumptions about the output distributions of the mechanism or use some new technique which is not reliant on the dense model theorem. This is due to that the blow-up of indistinguishability parameter in the dense model theorem used to derive our implication is tight for general distributions (see Section 5.4) and that our approximate translation from statistical distance to computational distance (Corollary 3) also is tight¹². The only known implication from IND-CDP to $\text{SIM}_{\forall\exists}$ -CDP goes via the dense model theorem, but to the best of our knowledge, it is plausible that some other proof technique could give a tighter implication.

Whereas all open regions of Table 1 are interesting in their own right, of particular interest are perhaps the region of all definitions with constant ε and non-negligible δ and the region of SIM-CDP and $\text{SIM}_{\forall\exists}$ -CDP for non-constant ε and any δ .

Open problem 2 *Are there any separations between the CDP definitions for arbitrary constants $\varepsilon > 0$ and $\delta \in (0, 1)$?*

Open problem 3 *Are (ε, δ) -SIM-CDP and (ε, δ) - $\text{SIM}_{\forall\exists}$ -CDP equivalent for non-constant ε and any $\delta < 1$?*

7.1 Other Open Directions

This work might be seen as an extension of the formal study of CDP from the setting analogous to pure SDP to that of approximate SDP, motivated by the prevalence of approximate SDP in the practice-oriented literature on DP. Therefore another intriguing direction is to do the same with respect to other versions of SDP that are commonly used, perhaps most pressingly *concentrated DP* [19, 11]. Interestingly there is, as far as we are aware, no work even proposing a computational version of concentrated DP.

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¹² The tightness, i.e. that $CD(X, Y) = \delta$ cannot imply $SD(Y, Z) = \delta'$ and $CD(X, Z) \leq \text{negl}$ for any δ' noticeably smaller than δ , follows immediately from that a computational distance is never larger than the corresponding statistical distance.

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A Deferred proofs

A.1 Proof of Proposition 3

Proof. The proof goes via the formulation of approximate SDP as smoothed SDP. Let $\mathcal{M}$ be an (ε, δ) -SDP mechanism. Set $\mathcal{S}(D) = \mathcal{M}(D)|E$ and $\mathcal{S}(D') = \mathcal{M}(D')|E'$. Then $\mathcal{S}(D)$ and $\mathcal{S}(D')$ are mutually $e^{-\varepsilon}$ -dense. What remains to show is that the statistical distance between $\mathcal{S}(D)$ and $\mathcal{M}(D)$ is at most δ (the computation for $\mathcal{S}(D')$ and $\mathcal{M}(D')$ is analogous). Assume without loss of generality that $\mathbb{P}(E) = 1 - \delta$. We have:

$$\begin{aligned}
SD(\mathcal{M}(D), S(D)) &= \frac{1}{2} \sum_{x \in \mathcal{R}} \left| \mathbb{P}(\mathcal{M}(D) = x) - \mathbb{P}(S(D) = x) \right| \\
&= \frac{1}{2} \sum_{x \in \mathcal{R}} \left| \mathbb{P}(\mathcal{M}(D) = x) - \mathbb{P}(\mathcal{M}(D) = x|E) \right| \\
&= \frac{1}{2} \sum_{x \in \mathcal{R}} \left| \mathbb{P}(\mathcal{M}(D) = x|E)\mathbb{P}(E) + \mathbb{P}(\mathcal{M}(D) = x|\neg E)\mathbb{P}(\neg E) \right. \\
&\quad \left. - \mathbb{P}(\mathcal{M}(D) = x|E) \right| \\
&= \frac{1}{2} \sum_{x \in \mathcal{R}} \left| (1 - \delta)\mathbb{P}(\mathcal{M}(D) = x|E) + \delta\mathbb{P}(\mathcal{M}(D) = x|\neg E) \right. \\
&\quad \left. - \mathbb{P}(\mathcal{M}(D) = x|E) \right| \\
&= \frac{1}{2} \sum_{x \in \mathcal{R}} \left| -\delta\mathbb{P}(\mathcal{M}(D) = x|E) + \delta\mathbb{P}(\mathcal{M}(D) = x|\neg E) \right| \\
&= \delta \frac{1}{2} \sum_{x \in \mathcal{R}} \left| -\mathbb{P}(\mathcal{M}(D) = x|E) + \mathbb{P}(\mathcal{M}(D) = x|\neg E) \right| \leq \delta.
\end{aligned}$$

Thus the proposition follows. $\square$

A.2 Proof of Proposition 4

Proof. That $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is (ε, δ) -SIM $_{\forall\exists}$ -CDP means that for every pair D, D' , $T \in \mathcal{T}_{poly}$ and sufficiently large κ there are $\mathcal{S}_\kappa(D), \mathcal{S}_\kappa(D')$ such that:

$$\begin{aligned}
\mathbb{P}(T(\mathcal{M}_\kappa(D)) = 1) &\leq \mathbb{P}(T(\mathcal{S}_\kappa(D)) = 1) + \text{negl}(\kappa) \\
&\leq e^{\varepsilon\kappa} \mathbb{P}(T(\mathcal{S}_\kappa(D')) = 1) + \delta_\kappa + \text{negl}(\kappa) \\
&\leq e^{\varepsilon\kappa} (\mathbb{P}(T(\mathcal{M}_\kappa(D')) = 1) + \text{negl}(\kappa)) + \delta_\kappa + \text{negl}(\kappa) \\
&= e^{\varepsilon\kappa} \mathbb{P}(T(\mathcal{M}_\kappa(D')) = 1) + \delta_\kappa + (e^{\varepsilon\kappa} + 1)\text{negl}(\kappa).
\end{aligned}$$

Fulfilling this inequality is precisely the criteria for $\mathcal{M}$ being $(\varepsilon, \delta + e^\varepsilon \text{negl})$ -IND-CDP and therefore the first part of the proposition follows.

Again, the second part of the proof follows that of Lemma 2 but in the computational setting. Set $\mathcal{R} = \mathcal{D} = \{0, 1\}$ and define $\mathcal{S}$ by $\mathbb{P}(\mathcal{S}_\kappa(D) = D) =$

$\frac{e^{\varepsilon\kappa}}{1+e^{\varepsilon\kappa}}(1-\delta_\kappa)+\delta_\kappa$. This is equivalent to $\mathcal{S}_\kappa(D)$ outputting D with probability δ_κ and otherwise outputting D with probability $\frac{e^{\varepsilon\kappa}}{1+e^{\varepsilon\kappa}}$, which means that $\mathcal{S}$ is (ε, δ) -SDP by the characterization of approximate SDP as smoothed DP. Now let $\mathcal{M}$ be defined by $\mathbb{P}(\mathcal{M}_\kappa(D) = D) := \mathbb{P}(\mathcal{S}_\kappa(D) = D) + \text{negl}(\kappa)$, for some negligible function negl . Clearly, $\mathcal{M}$ is (ε, δ) -SIM-CDP and thus also (ε, δ) -SIM $_{\forall\exists}$ -CDP. Thus, for any two adjacent datasets $D, D' \in \{0, 1\}$, the same set of inequalities as above gives that for all efficient tests T :

$$\mathbb{P}(T(\mathcal{M}_\kappa(D)) = 1) \leq e^{\varepsilon\kappa} \mathbb{P}(T(\mathcal{M}_\kappa(D')) = 1) + \delta_\kappa + (e^{\varepsilon\kappa} + 1)\text{negl}(\kappa). \quad (8)$$

That is, we have confirmed that $\mathcal{M}$ is indeed $(\varepsilon, \delta + e^{\varepsilon}\text{negl})$ -IND-CDP, as is implied by the first part of the proposition. What remains to prove now is that the mechanism is not IND-CDP for any smaller value choice of the second parameter. We do this by exhibiting an efficient binary test that such that Equation 8 is an equality rather than an inequality. Consider the test that simply outputs its input, i.e. $T(b) = b$ for $b \in \{0, 1\}$. Note that due to symmetry its complement exhibits probabilities analog to those below. For this test:

$$\mathbb{P}(\mathcal{M}_\kappa(D) = D) = \frac{e^{\varepsilon\kappa}}{1 + e^{\varepsilon\kappa}}(1 - \delta_\kappa) + \delta_\kappa + \text{negl}(\kappa), \quad (9)$$

and simultaneously, from the law of total probability,

$$\mathbb{P}(\mathcal{M}_\kappa(D') = D) = \frac{1}{1 + e^{\varepsilon\kappa}}(1 - \delta_\kappa) - \text{negl}(\kappa). \quad (10)$$

Therefore, by setting $D = 1, D' = 0$, we get

$$\begin{aligned} \mathbb{P}(\mathcal{M}_\kappa(D) = D) &= \mathbb{P}(\mathcal{M}_\kappa(1) = 1) = \frac{e^{\varepsilon\kappa}}{1 + e^{\varepsilon\kappa}}(1 - \delta_\kappa) + \delta_\kappa + \text{negl}(\kappa) \\ &= e^{\varepsilon\kappa} \frac{1}{1 + e^{\varepsilon\kappa}}(1 - \delta_\kappa) + \delta_\kappa + \text{negl}(\kappa) \\ &= e^{\varepsilon\kappa} \mathbb{P}(\mathcal{M}_\kappa(0) = 1) + e^{\varepsilon\kappa} \text{negl}(\kappa) + \delta_\kappa + \text{negl}(\kappa) \\ &= e^{\varepsilon\kappa} \mathbb{P}(\mathcal{M}_\kappa(D') = D) + \delta_\kappa + (e^{\varepsilon\kappa} + 1)\text{negl}(\kappa) \end{aligned}$$

That is, for the mechanism $\mathcal{M}$, there exists an efficient binary test such there are adjacent databases D, D' for which the inequality in Equation 8 is an equality. Therefore, $\mathcal{M}$ cannot be (ε, δ') -IND-CDP for any δ' noticeably smaller than $\delta + e^{\varepsilon}\text{negl}$ and thereby the second claim of the proposition follows.

□

A.3 Proof of Theorem 3

As discussed in the main body, the proof of Theorem 3 is exactly the same as [29]'s proof of the barrier in the case of $\delta = 0$ except for the one step where the CDP guarantee of the assumed mechanism is used. The outline of [29]'s proof is as follows:

1. The OWF used by $\mathcal{M}$ can be simulated by picking a random permutation (length-preserving function) $\mathcal{O}_{rand}$ and using that instead of $\mathcal{O}$, since with probability negligibly close to 1, a random function is one-way. Since the construction is fully black-box, the DP guarantee of the mechanism using the random function holds also against unbounded tests, as long as the test makes at most polynomially many calls to $\mathcal{O}_{rand}$. This is not a problem since in the proof below, the test is not given access to $\mathcal{O}_{rand}$.
2. The SDP mechanism is simply the same as $\mathcal{M}$ except for that it uses $\mathcal{O}_{rand}$ instead of the OWF $\mathcal{O}$. Therefore, by the bulletpoint above, the SDP mechanism can be shown to be SDP by essentially the same hybrid argument that shows an implication from $\text{SIM}_{\forall\exists}$ -CDP to IND-CDP.

Let F_κ be the set of length-preserving functions (permutations) from $\{0, 1\}^\kappa$ to $\{0, 1\}^\kappa$. The restriction to working over linearly many bits is without loss of generality, since we always assume that the elements in $\mathcal{R}_\kappa$ can be represented by polynomially many bits. The probability that $\mathcal{O}_{rand}$ is one-way was bounded in [25]:

Lemma 6 ([25]). *For sufficiently large κ , a function drawn uniformly at random from F_κ is one-way (in fact with inversion probability at most $2^{-\kappa/5}$ by circuits of size at most $2^{\kappa/5}$) with probability at least $1 - 2^{-2^{\kappa/2}}$.*

Proof (Theorem 3, [29]). Let $\mathcal{M}'_\kappa$ be the mechanism that draws $\mathcal{O}_{rand}$ uniformly from F_κ and then runs $\mathcal{M}_\kappa$ internally but with the calls to the OWF replaced by evaluations of $\mathcal{O}_{rand}$. Let E_{OW} denote the event that $\mathcal{O}_{rand}$ is one-way and $p_{OW}(\kappa)$ be the probability of E_{OW} when $\mathcal{O}_{rand}$ is drawn from F_κ . Then for any predicate test T on $\mathcal{R}_\kappa$ and any $D \in \mathcal{D}$, we have:

$$\begin{aligned}
\mathbb{P}(T(\mathcal{M}'_\kappa(D)) = 1) &= \mathbb{P}(T(\mathcal{M}'_\kappa(D)) = 1 | E_{OW}) p_{OW}(\kappa) \\
&\quad + \mathbb{P}(T(\mathcal{M}'_\kappa(D)) = 1 | \neg E_{OW}) (1 - p_{OW}(\kappa)) \\
&\leq \mathbb{P}(T(\mathcal{M}'_\kappa(D)) = 1 | E_{OW}) \cdot 1 + 1 \cdot (1 - p_{OW}(\kappa)) \\
&\leq e^{\varepsilon_\kappa} \mathbb{P}(T(\mathcal{M}'_\kappa(D')) = 1 | E_{OW}) + \delta_\kappa + \text{negl}(\kappa) + 2^{-2^{\kappa/2}} \\
&\leq e^{\varepsilon_\kappa} \mathbb{P}(T(\mathcal{M}'_\kappa(D')) = 1) + \delta_\kappa + \text{negl}(\kappa) + 2^{-2^{\kappa/2}}.
\end{aligned}$$

The second inequality follows from Lemma 6 above and that $\mathcal{M}$ is given only black-box access to the OWF. That is, $\mathcal{M}$ knows nothing about the OWF except it being one-way, thus $\mathcal{M}$ it is equivalent to $\mathcal{M}'$ when $\mathcal{O}_{rand}$ is one-way. Therefore $\mathcal{M}'$ in this case inherits the DP guarantee from $\mathcal{M}$ but in statistical form. Thus $\mathcal{M}'$ is $(\varepsilon, \delta + \text{negl})$ -SDP. Also, note that there is at most a negligible utility difference between $\mathcal{M}'$ and $\mathcal{M}^\mathcal{O}$, since they are identical in the case that $\mathcal{O}_{rand}$ is one-way and this happens with probability exponentially close to 1. $\square$

A.4 Proof sketch of Theorem 4

Proof (Theorem 4, sketch). Since $\mathcal{M}$ is (ε, δ) - $\text{SIM}_{\forall\exists}$ -CDP, there exists for every $D \in \mathcal{D}$ and large enough κ a distribution $\mathcal{S}_\kappa(D)$ and a negligible function negl_1

such that for all $T \in \mathcal{T}_{poly}$:

$$|\mathbb{P}(T(\mathcal{M}_\kappa(D)) = 1) - \mathbb{P}(T(\mathcal{S}_\kappa(D)) = 1)| \leq \text{negl}_1(\kappa).$$

Since $\mathcal{M}$ is efficient, it gives output representable by polynomially many bits. For clarity assume (w.l.o.g.) that output by all mechanisms are in fixed-point notation with κ bits of precision. That is, set $\mathcal{R}_\kappa := \{x \in \mathbb{R}^n | \exists j \in \mathbb{Z} : x = j \cdot 2^{-\kappa}\}$, for some constant integer n (independent of κ). For all $x \in \mathbb{R}$, $\kappa \in \mathbb{N}$, define $\lceil x \rceil_\kappa := \lceil x \cdot 2^\kappa \rceil 2^{-\kappa}$.

Now noisy versions of $\mathcal{M}_\kappa$ and $\mathcal{S}_\kappa(D)$, denoted $\hat{\mathcal{M}}_\kappa$ and $\hat{\mathcal{S}}_\kappa(D)$, are constructed by outputting a uniformly random value within a *box* (hypercube in $\mathcal{R}_\kappa$) centered around $\mathcal{M}_\kappa(D)$ (or $\mathcal{S}_\kappa(D)$). This box has L_∞ -radius equal to $c(\kappa) := \left\lceil \frac{1}{4^n \sqrt{\text{negl}_1(\kappa)}} \right\rceil_\kappa$. Since $\hat{\mathcal{S}}_\kappa(D)$ is simply post-processing of $\mathcal{S}_\kappa(D)$, the approximate density between $\mathcal{S}_\kappa(D)$ and $\mathcal{S}_\kappa(D')$ as guaranteed by $\mathcal{M}$ being $\text{SIM}_{\forall\exists}$ -CDP is unchanged. That is, for all predicate tests T on $\mathcal{R}_\kappa$:

$$\mathbb{P}(T(\hat{\mathcal{S}}_\kappa(D)) = 1) \leq e^{\varepsilon_\kappa} \mathbb{P}(T(\hat{\mathcal{S}}_\kappa(D')) = 1) + \delta_\kappa.$$

What now remains to show is that $\hat{\mathcal{S}}_\kappa(D)$ and $\hat{\mathcal{M}}_\kappa(D)$ are statistically close for all D and to plug all results about indistinguishability into the above inequality about density. It can be shown¹³ that for any predicate test T on $\mathcal{R}_\kappa$:

$$\begin{aligned} & |\mathbb{P}(T(\hat{\mathcal{S}}_\kappa(D)) = 1) - \mathbb{P}(T(\hat{\mathcal{M}}_\kappa(D)) = 1)| \\ & \leq \left(\frac{z(\kappa) + 1}{c(\kappa)} \right)^n \text{negl}_1(\kappa) + \frac{\text{err}(\kappa) + b(\kappa)c(\kappa)}{z(\kappa)} + \text{negl}_1(\kappa), \end{aligned}$$

where $z(\kappa) := \left\lceil \frac{1}{4^n \sqrt{\text{negl}_1(\kappa)}} \right\rceil_\kappa$ and $b(\kappa)$ is some polynomial. By substituting this into the inequality of approximate density, we get for any two adjacent D, D' :

$$\begin{aligned} \mathbb{P}(T(\hat{\mathcal{M}}_\kappa(D)) = 1) & \leq e^{\varepsilon_\kappa} \mathbb{P}(T(\hat{\mathcal{M}}_\kappa(D')) = 1) + \delta_\kappa \\ & + (e^{\varepsilon_\kappa} + 1) \left(\left(\frac{z(\kappa) + 1}{c(\kappa)} \right)^n \text{negl}_1(\kappa) + \frac{\text{err}(\kappa) + b(\kappa) \cdot c(\kappa)}{z(\kappa)} \right) \\ & + e^{\varepsilon_\kappa} \text{negl}_1(\kappa). \end{aligned}$$

Note that $\hat{\mathcal{M}}$ is indeed an explicit mechanism, since it is just post-processing of $\mathcal{M}$. For $\hat{\mathcal{M}}$ to be $(\varepsilon, \delta + \text{negl})$ -SDP, we need the second and third line of the inequality above to be negligible. Each term within the large paranthesis can be made negligible [29], but without any tighter bound, because the negligible functions are derived from the indistinguishability in $\text{SIM}_{\forall\exists}$ -CDP. Therefore, we know that $\hat{\mathcal{M}}$ is guaranteed to be $(\varepsilon, \delta + \text{negl})$ -SDP if and only if $\varepsilon \in O(\log(\kappa))$. Thus the theorem follows. $\square$

¹³ For details, see [29]. Note that these steps does not rely at all on the parameters ε, δ .